Cardiovascular Critical Care I

Stephanie Dwyer Kaluzna, Pharm.D., BCCP

University of Illinois Chicago College of Pharmacy
University of Illinois Hospital and Health Sciences System
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Learning Objectives

1. Develop an appropriate pharmacotherapeutic regimen based on a patient's hemodynamic status and objective cardiac findings.
2. Design a treatment plan for patients with cardiogenic shock.
3. Develop treatment plans for critically ill patients with cardiovascular diseases, including, but not limited to, coronary artery disease, arrhythmias, heart failure (HF), and valvular disease.
4. Recognize the utility of mechanical circulatory support and heart transplantation for patients with advanced HF and cardiogenic shock.

Abbreviations in This Chapter

ABG	Arterial blood gas
ACE	Angiotensin-converting enzyme
ACS	Acute coronary syndrome(s)
AF	Atrial fibrillation
ARB	Angiotensin receptor blocker
AV	Atrioventricular
BNP	Brain natriuretic peptide
CABG	Coronary artery bypass grafting
CAD	Coronary artery disease
CVP	Central venous pressure
ECG	Electrocardiography/electrocardiogram
ECHO	Echocardiography/echocardiogram
ECMO	Extracorporeal membrane oxygenation
HF	Heart failure
HOCM	Hypertrophic obstructive cardiomyopathy
ICU	Intensive care unit
LV	Left ventricle/ventricular
LVAD	Left ventricular assist device
LVEF	Left ventricular ejection fraction
MAP	Mean arterial pressure
MCS	Mechanical circulatory support
MI	Myocardial infarction
NSTEMI	Non–ST-segment elevation myocardial infarction
PCI	Percutaneous coronary intervention
PVR	Pulmonary vascular resistance
RV	Right ventricle/ventricular
SA	Sinoatrial
SCAI	The Society for Cardiovascular Angiography and Interventions
STEMI	ST-segment elevation myocardial infarction
SVR	Systemic vascular resistance
VAD	Ventricular assist device
VT	Ventricular tachycardia

Self-Assessment Questions

Answers and explanations to these questions may be found at the end of this chapter.

Questions 1–8 pertain to the following case.

A 58-year-old man (height 71 inches, weight 106 kg) was transferred to the intensive care unit (ICU) from an outlying hospital after 24 hours of progressively worsening chest pain and shortness of breath. He arrives on 6 L of oxygen by high-flow nasal cannula, with a heparin infusion at 16 units/kg/hour and dopamine infusion at 15 mcg/kg/minute. Notes show that he was bradycardic (heart rate of 50–58 beats/minute) and hypotensive (78/49–86/55 mm Hg) on presentation to the outlying hospital. His 12-lead electrocardiogram (ECG) at the outlying hospital showed ST-segment elevation in leads II, III, and aVF. Given his chest pain and ECG findings, aspirin 324 mg (chewed and swallowed), clopidogrel 600 mg, and morphine 4 mg intravenously once were administered before transfer. Notes also show that β-blockers and nitroglycerin were held because of bradycardia and hypotension.

His past medical history is significant for nonadherence, hypertension, diabetes, dyslipidemia, and heart failure with preserved ejection fraction (HF*p*EF), with a last-reported left ventricular ejection fraction (LVEF) of 65% 1 year ago. The patient reports that he currently takes no medications at home.

- Vital signs on transfer: blood pressure 87/52 mm Hg, heart rate 110 beats/minute, respiratory rate 19 breaths/minute, temperature 99.7°F (37.6°C)
- The team's physical assessment indicates ongoing distress, radiating chest pain 7/10, evidence of rales, absence of any cardiac murmurs, and presence of a right radial arterial line.
- A preexisting urinary catheter with 20 mL of urine in the reservoir is also present (the patient reports that his most recent void was yesterday morning). According to outlying hospital records, only 30 mL of urine was reported before time of transfer.
- His chest radiography reveals evidence of diffuse patchy opacities; however, the report indicates that an infiltrate cannot be ruled out.

- His serum chemistry panel results are as follows: sodium 132 mEq/L, potassium 4.2 mEq/L, chloride 102 mEq/L, carbon dioxide 22 mEq/L, blood urea nitrogen (BUN) 34 mg/dL, serum creatinine (SCr) 1.9 mg/dL, and glucose 163 mg/dL.
- Results of the complete blood cell count (CBC) are as follows: white blood cell count (WBC) 11.3 × 10^3 cells/mm^3, hemoglobin 10.9 g/dL, hematocrit 31.1%, and platelet count 213,000/mm^3.
- Additional laboratory values include troponin-T 3.9 ng/mL, aspartate aminotransferase (AST) 14 IU/L, alanine aminotransferase (ALT) 46 IU/L, hemoglobin A1C (A1C) 8.3%, and brain natriuretic peptide (BNP) of 1423 pg/mL.

1. Which is the most likely cause of this patient's admission and transfer?
   A. Septic shock caused by suspected pneumonia with subsequent myocardial depression and demand ischemia.
   B. Cardiogenic shock caused by acute on chronic decompensated systolic heart failure (HF).
   C. Cardiogenic shock caused by a suspected inferior ST-segment elevation myocardial infarction (STEMI).
   D. Cardiogenic shock caused by a suspected non–ST-segment elevation myocardial infarction (NSTEMI) affecting the lateral wall.

2. Given this patient's presentation, which coronary artery is most likely to be the culprit lesion?
   A. Left main coronary artery.
   B. Left anterior descending artery.
   C. Left circumflex coronary artery.
   D. Right coronary artery.

3. The interventional cardiologist who is evaluating the patient for potential revascularization asks the ICU team to place a central venous catheter. Which changes/interventions regarding this patient's hemodynamic support would be best to recommend?
   A. Increase dopamine to achieve a mean arterial pressure (MAP) greater than 65 mm Hg.
   B. Convert the patient to norepinephrine, and titrate the dose to achieve a MAP greater than 65 mm Hg while weaning off dopamine.
   C. Initiate milrinone at 0.375 mcg/kg/minute, and continue dopamine at 15 mcg/kg/minute.
   D. Administer 1000 mL of normal saline as a bolus because of low urine output.

4. The patient has been taken to the cardiac catheterization laboratory, and a "code blue" is called overhead for immediate emergency response to this patient's procedural area. On arrival, chest compressions have just been paused for defibrillation, and a single dose of epinephrine has been administered. The interventional team indicates that, when attempting visualization of the right coronary artery, the patient went into ventricular tachycardia (VT). The patient's telemetry monitor now shows sinus tachycardia with noted ectopy—heart rate 113 beats/minute and blood pressure 84/52 mm Hg. The cardiologist asks for recommendations for an antiarrhythmic because he is concerned about a VT recurrence, given the bigeminy on telemetry. Which agent would be best to recommend?
   A. Lidocaine 100 mg intravenous push for 2–3 minutes, followed by an infusion at 1 mg/minute.
   B. Amiodarone 300 mg intravenous push for less than 1 minute.
   C. Metoprolol 10 mg intravenous push for 1–2 minutes.
   D. Diltiazem 20 mg intravenous push for 2 minutes, followed by a continuous infusion at 5 mg/hour and titrated to maintain a heart rate less than 110 beats/minute.

5. The patient returns to the ICU after his left heart catheterization, which was performed through the femoral artery. In addition to his acute decompensation, which major procedural complication is of greatest concern during the next 12 hours?
   A. Bleeding (particularly retroperitoneal bleeding).
   B. Dissection/rupture.
   C. Stent thrombosis.
   D. Papillary muscle rupture.

6. The patient returns to the ICU with a pulmonary artery catheter in place and is currently receiving dopamine at 12 mcg/kg/minute and norepinephrine at 0.08 mcg/kg/minute with heart rate 108 beats/minute, blood pressure 82/51 mm Hg, cardiac index 2.0, central venous pressure (CVP) 26 mm Hg, and pulmonary artery pressure 49/21 mm Hg. The physician is concerned about right ventricular dysfunction in the setting of shock and approaches you for a recommendation to increase blood pressure (BP). Ideally, the physician would prefer to wean dopamine and minimize further increases in pulmonary vascular resistance (PVR) because of the presence of pulmonary hypertension. Which strategy would be best to recommend?
   A. Keep the current infusions, and reevaluate later.
   B. Initiate phenylephrine at 1 mcg/kg/minute, and wean the dopamine off if MAP is greater than 65 mm Hg.
   C. Initiate vasopressin at 0.04 unit/minute, and wean the dopamine off if MAP is greater than 65 mm Hg.
   D. Administer a 1-L bolus of normal saline, and wean the dopamine off if MAP is greater than 65 mm Hg.

7. Hours later, this patient goes into atrial fibrillation (AF) with a heart rate of 126 beats/minute; however, the patient's blood pressure remains 86/56 mm Hg according to the regimen selected in the previous question. Which agent would you most likely administer to manage the patient's AF?
   A. Amiodarone 150 mg intravenous push, followed by a continuous infusion at 1 mg/minute.
   B. Amiodarone 150 mg intravenous infusion for 10 minutes, followed by a continuous infusion at 1 mg/minute.
   C. Metoprolol 5 mg intravenous push once, followed by 5 mg intravenous push every 6 hours.
   D. Diltiazem 20 mg intravenous push, followed by a continuous infusion at 5 mg/hour and titrated to maintain a heart rate of less than 110 beats/minute.

8. Which medication-related quality metric would not require documentation of contraindications based on this patient's clinical presentation (acute MI with preserved LVEF)?
   A. Aspirin contraindication.
   B. Statin contraindication.
   C. β-Blocker contraindication.
   D. Angiotensin-converting enzyme (ACE) inhibitor/angiotensin receptor blocker (ARB) contraindication.

BPS Critical Care Pharmacy Examination Content Outline

This chapter covers the following sections of the Critical Care Pharmacy Examination Content Outline:

1. Domain 1: Critical Care
   a. Task A: 1, 2
   b. Task B: 1
2. Domain 2: Therapeutics and Patient Management
   a. Task A: 1, 2
   b. Task B: 2
   c. Task C: 2, 3
3. Domain 3: Professional Practice
   a. Task B: 1

I. CARDIOVASCULAR FUNDAMENTALS OVERVIEW

A. The myocardium has five functions and distinctive properties:
   1. Chronotropy: Ability to generate an electrical impulse at an intrinsic rate
   2. Dromotropy: The speed and ability to facilitate electrical impulse conduction
   3. Inotropy: Ability to contract in relation to a given preload, afterload, and heart rate
   4. Bathmotropy: Demonstration of an intrinsic excitatory threshold
   5. Lusitropy: Relaxation of the myocardium, independent from termination of contraction
   6. Other related terms:
      a. Preload: Volume of blood (represented in most cases by a pressure) in a ventricular cavity at the end of diastole imparting stretch on a resting myocardial sarcomere
      b. Afterload: The pressure that a *ventricle* must overcome to generate cardiac output. The greater the afterload (vascular resistance or impedance), the greater the amount of energy and force required to enable ejection of blood from a ventricle.

B. Coronary Artery Circulation
   1. Myocardial perfusion occurs through the coronary arteries during diastole.
   2. Coronary artery anatomy and perfusion are not the same in everyone.
   3. ECG abnormalities, hemodynamic assessment, and patient symptoms may assist with coronary artery disease (CAD) localization.
   4. Circulatory dominance:
      a. Right dominant: Posterior descending artery and atrioventricular (AV) nodal artery arise from the right coronary artery (85% of the population).
      b. Left dominant: Posterior descending artery arises from the circumflex artery (8% of the population).
      c. Codominant: Posterior descending artery arises from branches of the circumflex and the right coronary artery (7% of the population).
      d. Other notable variations: The sinoatrial (SA) node may have variation in the vessels that supply it; it is most commonly perfused by the right coronary artery (about 70%), circumflex (about 25%), or right coronary artery and circumflex (about 5%).

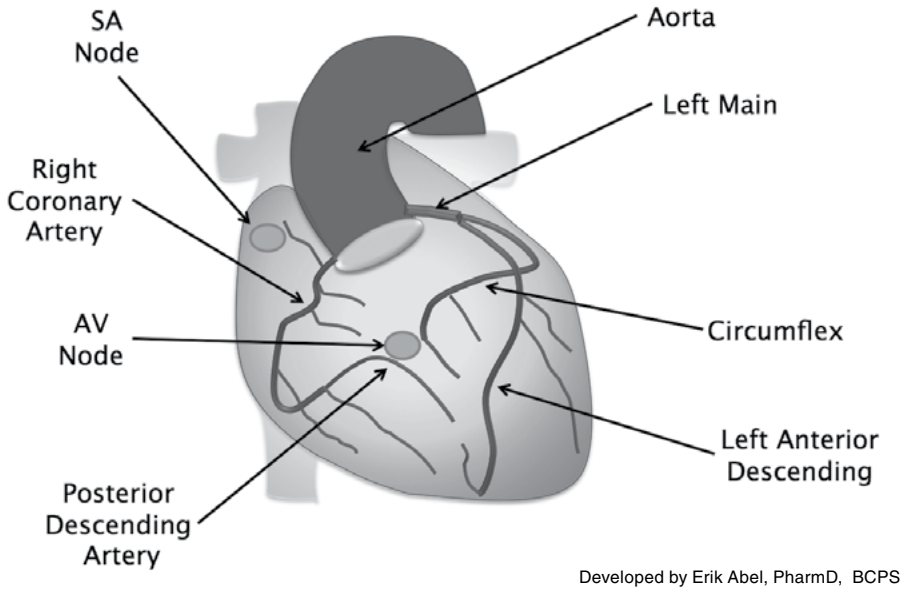
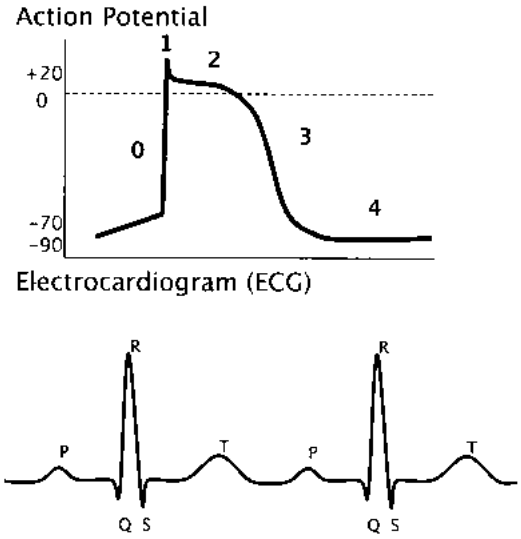


Figure 1. Coronary artery circulation.

C. Cardiac Anatomy in Relation to the ECG (Figure 2)
   1. A single lead of an ECG tracing is a summative representation of the action potentials occurring from a single cell, facilitating myocardial conduction, contraction, and relaxation.
   2. A 12-lead ECG can provide a geographic representation of conduction within the myocardial tissue. Conduction abnormalities within specified leads may indicate perfusion defects in some clinical scenarios (i.e., acute coronary syndromes [ACS]).



Phase	Electrolyte movement	Conduction change
0	Na^+ influx into cell	Depolarization
1	K^+ and Cl^- out of cell	Repolarization
2	Ca^{2+} into cell and K^+ out of cell	Plateau
3	K+ out of cell	Repolarization
4	Resting Membrane Potential	

ECG point	Representation
P	Sinoatrial (SA) node impulse generation/atrial depolarization
PR interval	Duration for conduction to occur from the SA node through the AV node, bundle of His, bundle branches and Purkinje fibers
QRS	Concurrent atrial repolarization and ventricular depolarization
ST segment	Ventricular refractory period
T	Ventricular repolarization
QT interval	Duration for ventricular repolarization
RR interval	Duration between ventricular depolarizations (represents heart rate)
U	Further ventricular depolarizations not listed or commonly visualized; may be more prominent in hypothermia, bradycardia, or severe electrolyte depletions

12 — Lead Electrocardiogram (ECG)

Limb leads				Precordial Leads			
I	Anterolateral (Circumflex)	aVR		V1	Anteroseptal (LAD)	V_4	Anteroapical (Distal LAD)
II	Inferior (RCA)	aVL	Anterolateral (Circumflex)	V2	Anteroseptal (LAD) Posterior (RCA)	V_5	Anterolateral (Circumflex)
III	Inferior (RCA)	aVF	Inferior (RCA)	V3	Anteroapical (Distal LAD) Posterior (RCA)	V_6	Anterolateral (Circumflex)

adapted from ACLS instructor book

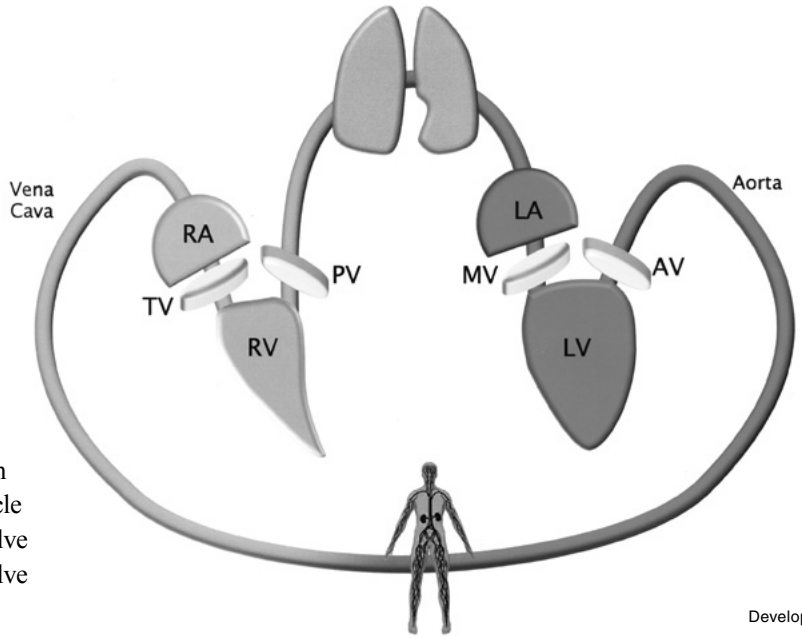
Figure 2. Cardiac Anatomy in relation to the ECG.

II. HEMODYNAMIC MANAGEMENT AND THE HEART

A. Hemodynamic Assessment Relies on the Clinician's Ability to:
   1. Understand cardiovascular circulation (Figure 3) and pathophysiology contributing to the hemodynamic variables in isolation and in the context of other changes and conditions.
   2. Account for hemodynamic trends in view of end-organ function and use surrogates of oxygen delivery.
   3. Understand the roles of hemodynamic tools, limitations in use, and interpretation of the devices/technology.
   4. Identify appropriate therapeutic targets (Figure 4), and apply the pharmacologic/pharmacodynamic principles (Table 1) to initiate, modify, or discontinue therapy depending on clinical response.

B. Advantages and Disadvantages of Invasive Hemodynamic Monitoring for Assessing Cardiac Output and/or Volume Status
   1. Hemodynamic monitoring techniques facilitate diagnosis
   2. Therapy guided by clinical assessment and a pulmonary artery (PA) catheter compared with clinical assessment alone for heart failure (HF) did not result in differences in overall mortality or hospitalizations. However, more anticipated adverse events (i.e., infection, bleeding) were seen with PA catheter-guided care (JAMA 2005;294:1625-33).
   3. However, for patients with SCAI Stages C–E cardiogenic shock, using PA catheter-derived hemodynamic data has been associated with improved mortality (J Am Coll Cardiol HF 2020;8:903-13).
   4. Diligent use and skilled interpretation of these technologies, together with patient assessment, can be used to guide therapeutic interventions.
   5. See comparisons of invasive and minimally invasive hemodynamic monitoring devices (Table 2 in the Shock Syndromes I chapter).

C. Pharmacologic Support in Cardiovascular Critical Illness (see Table 1)



RA = Right Atrium
RV = Right Ventricle
TV = Tricuspid Valve
PV = Pulmonic Valve

Left Atrium = LA
Left Ventricle = LV
Mitral Valve = MV
Aortic Valve = AV

Developed by Erik Abel, PharmD, BCPS

Figure 3. Cardiovascular systemic circulation.

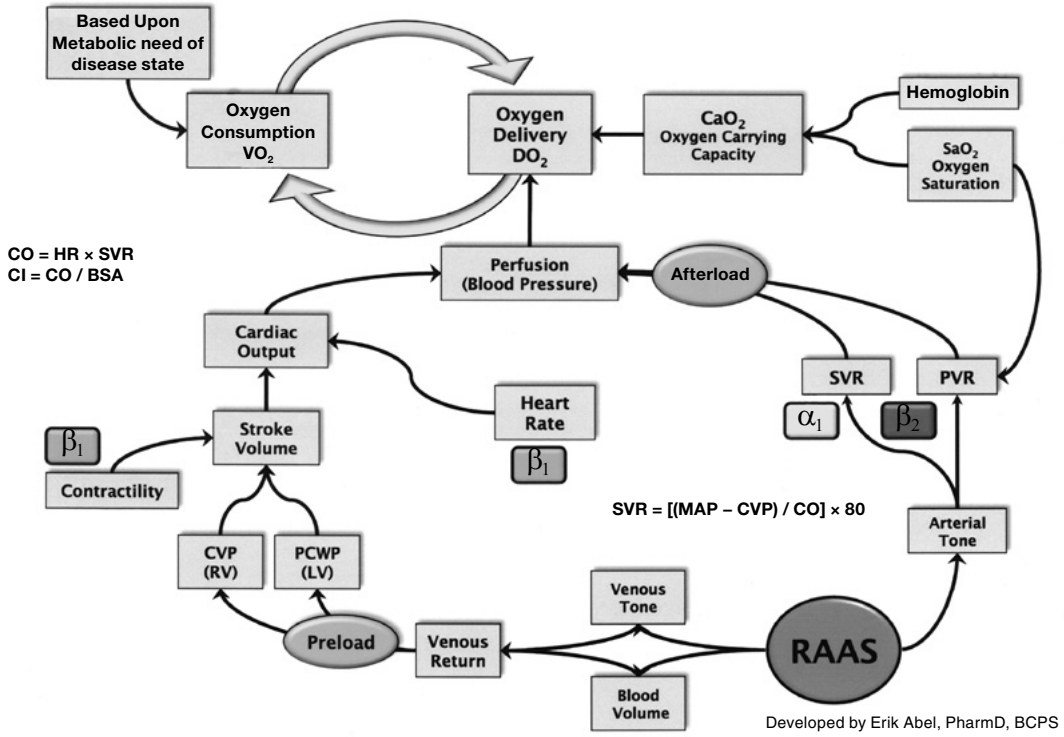


Figure 4. Integrated model of the hemodynamic variables and therapeutic targets.

BSA = body surface area; CI = cardiac index; CO = cardiac output; CVP = central venous pressure; HR = heart rate; MAP = mean arterial pressure; SVR = systemic vascular resistance.

Table 1. Pharmacologic Support in Cardiovascular Critical Illness

Vasopressors	Dopa	α_1	β_1	β_2	Other Mechanism	HR	CVP	CO	SVR	PVR
Dopamine[a]	+++++	+++	++++	++		↑	↔	↑	↔ or ↑	↔ or ↑
Epinephrine		++++	++++	+++		↑	↔ or ↑	↑	↑	↑
Norepinephrine		+++++	+++	++		↑ or ↔	↑	↑ or ↔	↑	↑
Phenylephrine		+++++				↔ or ↓	↑	↔ or ↓	↑	↑
Vasopressin	N/A “α_1-like effects”				V_1 and V_2 agonism	↔ or ↓	↑	↔ or ↓	↑	↔ or ↓
Angiotensin II	N/A “α_1-like effects”				AT1 receptor agonism	↔ or ↓	↑	↔ or ↓	↑	↔ or ↑
Inotropes										
Dobutamine		+	++++	++		↑	↔ or ↓	↑	↔ or ↓	↔ or ↓
Isoproterenol			+++++	+++++		↑	↔ or ↓	↑	↔ or ↓	↔ or ↓
Milrinone[b]	N/A “B_1 and β_2-like effects”				PDE_3 inhibition	↑	↓	↑	↓	↓

Table 1. Pharmacologic Support in Cardiovascular Critical Illness *(continued)*

Vasoactives	Dopa	α_1	β_1	β_2	Other Mechanism	HR	CVP	CO	SVR	PVR
Vasodilators										
Nitroglycerin					cGMP	↔ or ↑	↓	↔ or ↑	↔ or ↓	↔ or ↓
Nitroprusside					cGMP	↔ or ↑	↓	↔ or ↑	↓	↓
Nitric oxide (inhaled)					cGMP	N/A	↔	↔ or ↑	↔	↓
Epoprostenol (inhaled)					cAMP	N/A	↔	↔ or ↑	↔	↓

[a]High doses associated with increasing α_1 activity.

[b]Normal half-life is 2.5 hours, but drug is eliminated renally. Loading dose rarely used in routine management because of hypotension.

AMP = cyclic adenosine monophosphate; AT1 = angiotensin II type 1; cGMP = cyclic guanosine monophosphate; CVP = central venous pressure; HR = heart rate; N/A = not applicable; PDE3 = phosphodiesterase type 3; PVR = pulmonary vascular resistance; SVR = systemic vascular resistance. Information from: Br J Pharmacol 2012;165:2015-33; J Am Coll Cardiol 2014;63:2069-78; Pathophysiol Heart Dis 2011:xiv; Circulation 2008;118:1047-56; Crit Care Med 2014:xix; J Pharm Pract 2011;24:44-60.

III. CARDIOGENIC SHOCK

A. Characterized by three hallmarks:
   1. Sustained hypotension unresponsive to fluid administration alone (systolic blood pressure [SBP] less than 90 mm Hg for at least 30 minutes) (Cardiol Clin 2013;4:567-80)
   2. Evidence of myocardial dysfunction with reduced cardiac index (less than 2.2 L/minute/m^2)
   3. Signs and symptoms of malperfusion in the setting of volume overload and elevated cardiac filling pressures (e.g., pulmonary capillary wedge pressure greater than 18 mm Hg)

B. Society for Cardiovascular Angiography and Interventions (SCAI) Classifications for Cardiogenic Shock (Catheter Cardiovasc Interv 2019;94:29-37):
   1. Stage A – at risk. These patients are not in shock but are at risk of future development.
   2. Stage B – beginning. These patients have clinical evidence of relative hypotension or tachycardia without hypoperfusion.
   3. Stage C – classic. These patients have evidence of hypotension and hypoperfusion requiring intervention, including vasoactive therapies and mechanical circulatory support.
   4. Stage D – deteriorating. These patients are those in stage C who are getting worse despite attempts to restore optimal hemodynamics.
   5. Stage E – extremis. These patients are experiencing cardiac arrest with ongoing cardiopulmonary resuscitation or extracorporeal membrane oxygenation (ECMO).

C. Signs and Symptoms: See Heart Failure section.

D. Epidemiology: Without appropriate diagnosis and management, in-hospital mortality rates as high as 60% have been described (Semin Respir Crit Care Med 2011;32:598-606).

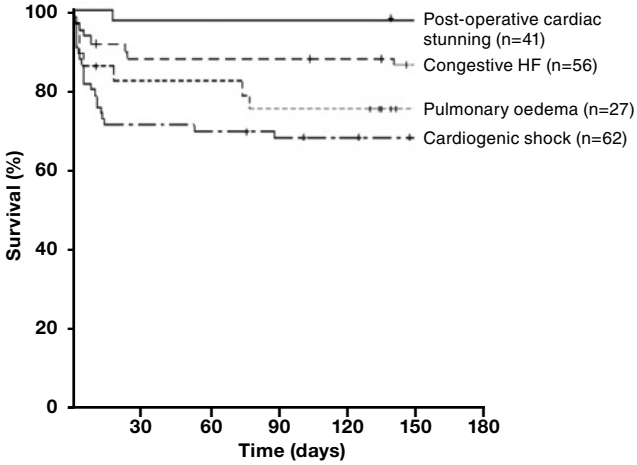


Figure 5. Survival rates of ICU patients over time presenting with different acute heart failure syndromes (Crit Care 2010;14:201).

E. Etiology
   1. Usually caused by left ventricular (LV) failure secondary to an acute MI, but a list of other potential causes can be found in Box 1
   2. May be multifactorial, including one or more of the causative etiologies; however, can also coexist with other types of shock syndromes

Box 1. Potential Causes of Cardiogenic Shock (Critical Care Medicine 2014:xix; Cardiol Clin 2013;31:567-80, viii; Cardiol Clin 2013;31:519-31, vii-viii; Semin Respir Crit Care Med 2011;32:598-606)

LV Failure
- Large MI
- Small MI with preexisting systolic HF
- Reinfarction
- Septic shock with severe myocardial depression

RV Failure
- RV infarction
- End-stage pulmonary hypertension

Acute Mechanical Dysfunction
- Papillary muscle rupture or chordal rupture with subsequent severe mitral regurgitation
- Free-wall rupture
- Ventricular septal rupture
- Cardiac tamponade

Cardiomyopathy
- End-stage HF
- Myocarditis
- Peripartum cardiomyopathy
- Left ventricular outflow tract obstruction
- Stress-induced cardiomyopathy (i.e., Takotsubo)

Valvular Disease
- Acute aortic regurgitation
- Ischemic mitral regurgitation
- Aortic or mitral stenosis with tachyarrhythmia or other condition causing decompensation
- Infectious endocarditis
- Prosthetic valve dysfunction or thrombosis

Arrhythmias
- Atrial tachyarrhythmias
- Ventricular tachyarrhythmias
- Bradycardia

Other Conditions
- Prolonged cardiopulmonary bypass and/or coronary air embolus
- Cardiac trauma (blunt or penetrating)
- Heart transplant rejection
- Pulmonary embolism
- Medical nonadherence

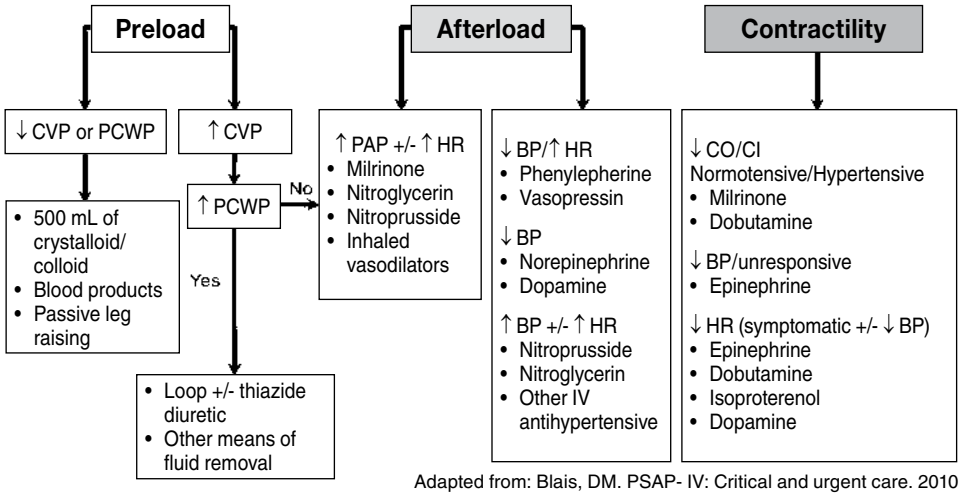
HF = heart failure; LV = left ventricular; MI = myocardial infarction; RV = right ventricular.

F. Resuscitation/Treatment
   1. Treatment largely depends on managing underlying chronic or acute cardiovascular disease(s) outlined in Box 1, with consideration given to chronicity of clinical changes before onset of shock.
   2. Means of management will be primarily discussed under the Major Contributing Etiologies headings in sections IV–VII.
   3. Hemodynamic management and pharmacotherapeutic considerations
      a. Because cardiogenic shock (CS) may have different underlying, contributing etiologies, hemodynamic management requires careful interpretation of clinical values. Treatment strategies could be devised according to the algorithm in Figure 6.

b. Milrinone as Compared with Dobutamine in the Treatment of Cardiogenic Shock (DOREMI) (N Engl J Med 2021;385:516-25)
   i. First prospective trial studying milrinone and dobutamine for CS management (n=192) during the index hospitalization
      (a) Median ejection fraction was 25%, and around two-thirds of patients had ischemic cardiomyopathy. Most patients (90%) fell into SCAI CS classes C (classic) and D (deteriorating).
      (b) Milrinone or dobutamine was initiated at a dose determined by a standard dosing scale ranging from 1 to 5, corresponding to 2.5, 5, 7.5, 10, and greater than 10 mcg/kg/minute for dobutamine and 0.125, 0.25, 0.5, and greater than 0.5 mcg/kg/minute for milrinone. Inotropes were adjusted in a blinded fashion.
   ii. Composite primary outcome (in-hospital mortality from any cause, resuscitated cardiac arrest, receipt of cardiac transplantation, or mechanical circulatory support, nonfatal MI, transient ischemic attack or stroke, or initiation of renal replacement therapy) did not differ between the two groups. Of importance, this study highlights the persisting high mortality rates in patients with CS (around 40%).
   iii. Both dobutamine and milrinone had high rates of arrhythmias leading to intervention, and there was no difference in sustained hypotension or vasopressor requirements for either medication.

c. Sepsis Occurrence in Acutely Ill Patients (SOAP) II landmark trial (N Engl J Med 2010;362:779-89)
   i. International multicenter trial (n=1679) that compared the 28-day mortality of norepinephrine with that of dopamine in the management of shock
      (a) Included if signs of malperfusion and MAP less than 70 mm Hg or SBP less than 100 mm Hg, despite adequate fluid challenge (at least 1000 mL of crystalloids or 500 mL of colloids unless the CVP was greater than 12 mm Hg or the pulmonary artery occlusion pressure was greater than 14 mm Hg)
      (b) Allowed titration to maximums of norepinephrine 0.19 mcg/kg/minute versus dopamine 20 mcg/kg/minute, after which open-label norepinephrine was allowed (open-label norepinephrine doses did not exceed 1.1 mcg/kg/minute during the study period)
      (c) Epinephrine and vasopressin were permitted as rescue agents (similar use in both groups).
      (d) Dobutamine use was greater in the norepinephrine group (19.4% vs. 14.8%).
   ii. Showed that dopamine was associated with a significantly higher mortality rate than was norepinephrine with or without dobutamine in patients with CS in subgroup analysis
   iii. Dopamine was associated with more adverse events than norepinephrine in the management of all shock subtypes, predominantly driven by the incidence of arrhythmias (24.1% vs. 12.1%, $p<0.001$). Tachyarrhythmias with dopamine had the greatest incidence within the first 36 hours after randomization.

d. Study Comparing the Efficacy and Tolerability of Epinephrine and Norepinephrine in CS [OptimaCC] (J Am Coll Cardiol 2018;72:173-82)
   i. Prospective, double-blind, multicenter, randomized study (n=57) that compared cardiac index evolution and refractory CS between epinephrine and norepinephrine in patients with CS post-MI
   ii. Trial showed that cardiac index evolution did not differ between groups, though the rate of refractory CS was significantly higher in the epinephrine arm than in the norepinephrine arms (37% vs. 7%).
   iii. Tachycardia and lactic acidosis were also significantly increased in the epinephrine group.



Adapted from: Blais, DM. PSAP- IV: Critical and urgent care. 2010

Figure 6. Shock management and treatment considerations based on hemodynamic indices.

Patient Case

Questions 1–8 pertain to the following case.
J.M. is a 68-year-old man with a history of CAD including STEMI 6 months ago with placement of two drug-eluting stents, type 2 diabetes, hypertension, dyslipidemia, gastroesophageal reflux disease, obstructive sleep apnea, frequent epistaxis, ischemic cardiomyopathy (LVEF 30%–35%), and moderate to severe mitral regurgitation. He is admitted to the cardiac intensive care unit for severe shortness of breath and altered mental status, and he is currently on continuous positive airway pressure (CPAP). He has gained 13 kg during the past 2 weeks (now weighs 121 kg) and has had decreased urine output, despite having had his diuretic dose increased. Home medications include the following: aspirin 81 mg once daily, ticagrelor 90 mg every 12 hours, pantoprazole 40 mg once daily, atorvastatin 40 mg once daily, insulin glargine 10 units every night, metformin 500 mg every 12 hours, furosemide 40 mg twice daily, and potassium chloride 20 mEq twice daily. His wife states that he was taking clopidogrel until 1 month ago, at which time he was given ticagrelor samples from his primary care physician because of cost concerns. A physical examination reveals rales throughout his lung fields. He is afebrile, anxious, and alert.

1. Given J.M.'s comorbidities, some drugs or drug classes have proven mortality benefits. Which medications could be added or modified to the patient's home medication profile to further slow disease progression and improve mortality?
   A. Amlodipine, clopidogrel, and sitagliptin
   B. Spironolactone, lisinopril, and carvedilol
   C. Pravastatin, amlodipine, and aspirin
   D. Prasugrel, sildenafil, and atenolol

2. Given the patient's presentation, which group of diagnostic tests would be most helpful to guide your recommendations to the team for J.M.'s current management? (Assume that a basic metabolic panel [Chem 7], a pulse oximetry, and a capillary blood glucose have already been completed.)
   A. Urine culture, respiratory culture, lactate, and procalcitonin
   B. Chest radiography, arterial blood gas (ABG), liver function tests, and serial troponins
   C. Chest radiography, echocardiogram (ECHO), lactate, and BNP
   D. Chest radiography, arterial line, ABG, and lactate

Patient Case *(contined)*

3. All tests previously mentioned have been ordered, and the following results are available. J.M.'s blood pressure is 90/56 mm Hg (MAP 67 mm Hg), and his heart rate is 56 beats/minute. A 12-lead ECG showed normal sinus rhythm without evidence of acute ST-T changes.
   - Chest radiography reveals diffuse patchy opacities; however, infiltrate cannot be ruled out; lines are all in appropriate positions.
   - Serum chemistry panel results are as follows: sodium 126 mEq/L, potassium 4.8 mEq/L, chloride 102 mEq/L, carbon dioxide 21 mEq/L, BUN 32 mg/dL, SCr 1.6 mg/dL, and glucose 134 mg/dL.
   - Results of the CBC are as follows: WBC 9.8×10^3 cells/mm^3, hemoglobin 11.1 g/dL, hematocrit 32.6%, and platelet count 173,000/mm^3.
   - Additional laboratory values include the following: troponin 0.9 ng/mL, AST 114 IU/L, ALT 102 IU/L, and BNP 1936 pg/mL.
   - Invasive hemodynamic variables include CVP 28 mm Hg, pulmonary artery pressures 46/22 mm Hg, cardiac index 1.8 L/minute/m^2, and central venous oxygen saturation ($ScvO_2$) 53%; pulmonary artery occlusion pressure is not yet available.
   - ABG results are as follows: pH 7.36, partial pressure of oxygen (Po_2) 93.7, partial pressure of carbon dioxide (Pco_2) 43.2, bicarbonate 23.9, oxygen (O_2) saturation 89%, and lactate 6.9.
   - ECHO results are pending.

   The patient's physical examination reveals that his extremities are cold to the touch, and his capillary refill is poor. The team has ordered furosemide 80 mg intravenously once and would like to initiate a vasopressor or inotrope for this patient. Which would be best to recommend at this time?

   A. Norepinephrine 0.08 mcg/kg/minute
   B. Epinephrine 0.08 mcg/kg/minute
   C. Milrinone 0.75 mcg/kg/minute
   D. Dobutamine 5 mcg/kg/minute

IV. ACUTE CORONARY SYNDROMES

A. Pathophysiology
   1. Manifestation of prolonged cessation of oxygenated blood supply to a portion of the myocardium that is most commonly caused by an acute thrombus at the site of coronary atherosclerotic stenosis leading to local or regional myocardial ischemia and necrosis
   2. Other disease states leading to an elevated myocardial oxygen demand with a concurrent inability to meet such demands may result in a scenario where "demand ischemia (type II NSTEMI)" is considered versus a diagnosis of ACS.

B. Presentation and Diagnosis (Circulation 2014;130:e344-426; Circulation 2013;127:e362-425) – Although other cardiac enzymes assays are available for clinical use, cardiac troponins are usually used as sensitive markers indicative of myocardial necrosis, with a negative troponin conferring a greater than 95% negative predictive value for MI. Troponin is eliminated renally, so patients with end-stage renal disease often have elevations in cardiac troponin without other evidence of myocardial ischemia. Therefore, other clinical signs and symptoms of acute coronary syndromes as well as the rate of troponin increase should be assessed.

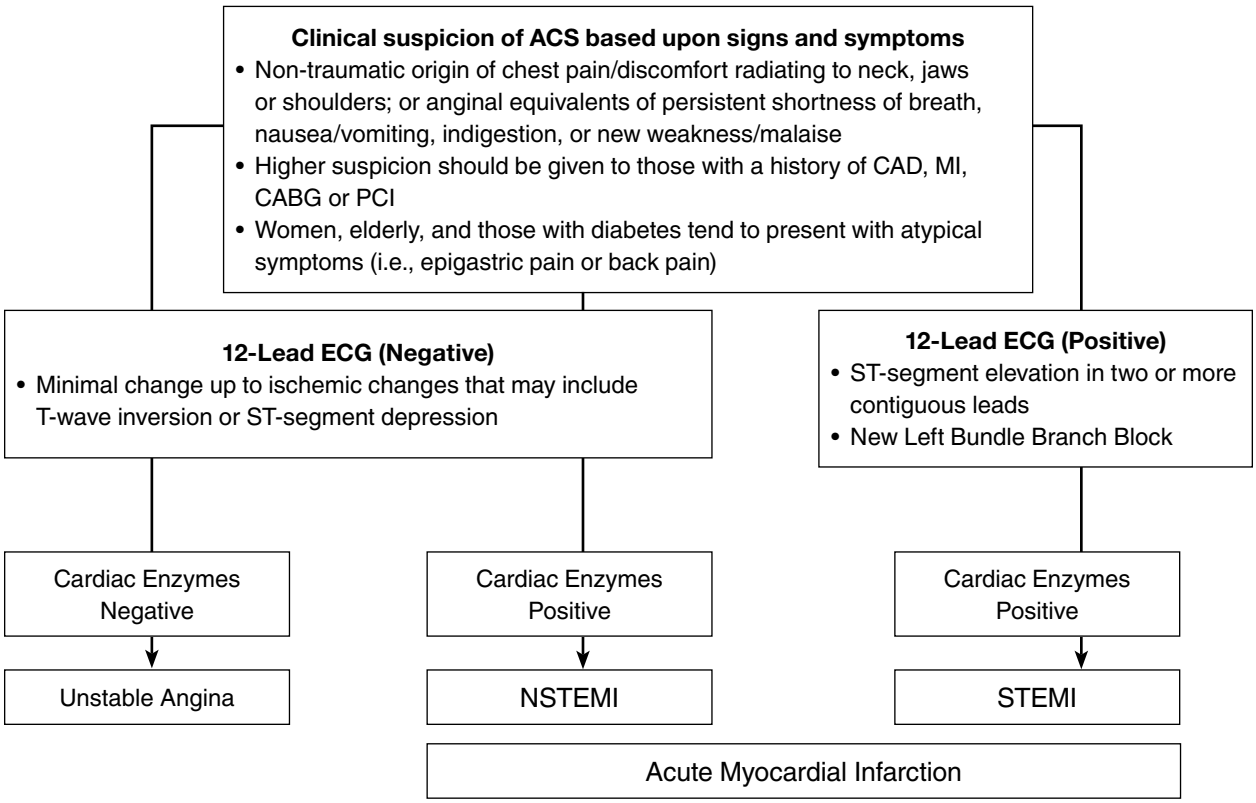


Figure 7. Acute coronary syndrome (ACS) presentation and diagnosis.

Box 2. Causes of Troponin Elevation (Heart 2006;92:987-93)

Acute decompensated heart failure	Early post-cardiac surgery
Acute MI	Heart transplantation
Acute pulmonary embolism	Myocarditis
Aortic stenosis	Pericarditis
Cardiac amyloidosis	Post-PCI
Cardiotoxic chemotherapy	Rhabdomyolysis
Chest compressions	Sepsis
Chest wall trauma or compressions	Severe strenuous exercise
Chronic heart failure	Tachyarrhythmia
Direct current cardioversion/defibrillation	Type A aortic dissection

PCI = percutaneous coronary intervention.

C. Acute Management of MI

Table 2. Acute Management (Circulation 2014;130:e344-426; Circulation 2013;127:e362-425)

	STEMI	NSTEMI/ Unstable Angina
Goals of care	Reperfusion therapy as soon as possible • Primary PCI preferred if it can be performed within 90 min of first medical contact • If primary PCI is not immediately available, and the delay from hospital presentation to PCI is anticipated to be >120 min; fibrinolytic should be administered within 30 min of presentation unless contraindications exist • Surgical revascularization may be indicated, depending on severity of CAD, complexity of anatomy, or development of other complications	Prevent total occlusion of the vessel • Decision and need for revascularization (PCI or surgery) vs. medical management should be made on the basis of risk stratification, symptom resolution, and indicators of ongoing myocardial damage/ischemia

CAD = coronary artery disease.

Table 3. Predominant Interventions on Presentation/Onset for Stabilization – Any ACS (Circulation 2014;130:e344-426; Circulation 2013;127:e362-425)

Predominant interventions on presentation/onset for stabilization – Any ACS	
Morphine or other narcotic analgesic	• Provides analgesia, and decreases pain-induced sympathetic/adrenergic tone • Morphine commonly used because it may reduce myocardial and microvascular damage by inducing vasodilation and mediate some degree of preload reduction; therefore, decreasing myocardial oxygen demand • Morphine is also an independent predictor of high residual platelet reactivity in ticagrelor/prasugrel use (may impair absorption because of delayed gastric emptying) (Am Heart J 2014;6:909-14) • Morphine now carries a class IIb-B recommendation and may be not be favored more than other narcotic analgesics, given that at least two large trials have identified an association between morphine administration and risk of death (N Engl J Med 2014;371:1016-27; Am Heart J 2005;149:1043-9; J Am Coll Cardio. 2014;64:e139-228)
Oxygen	• Can help attenuate anginal pain secondary to tissue hypoxia • Current evidence shows no benefit for supplemental oxygen (and potential harm) in patients with normal Sa_{O_2} (N Engl J Med 2017;13:1240-9; Circulation 2015;24:2143-50) • Supplemental oxygen considered if Sa_{O_2} < 90%, respiratory distress, or other high-risk features for hypoxemia
Nitrates	• Can facilitate coronary vasodilation and may also be helpful in scenarios of severe cardiogenic pulmonary edema caused by increased venous capacitance • Nitroglycerin (NTG) 0.4 mg sublingually every 5 minutes for total of THREE doses; afterward, assess need for IV NTG • IV NTG is indicated in the first 48 hr after onset of ischemia, HF, or hypertension but should not preclude therapy such as β-blockers or ACE inhibitors when indicated **Should <u>NOT</u> be administered:** • If SBP < 90 mm Hg OR if SBP is > 30 mm Hg below baseline • If severe bradycardia (including heart block) with HR ≤ 50 beats/min • If tachycardia (HR ≥ 100 beats/min) in the absence of symptomatic HF • RV infarction, as evidenced by ischemic changes in inferior leads on ECG • If patient has received an oral phosphodiesterase inhibitor or riociguat within the past 24–48 hr

Table 3. Predominant Interventions on Presentation/Onset for Stabilization – Any ACS (Circulation 2014;130:e344-426; Circulation 2013;127:e362-425) *(continued)*

Predominant interventions on presentation/onset for stabilization – Any ACS	
Aspirin	• Inhibits platelet activation • Four aspirin 81 mg each (324 mg total) or one 325 mg tablet (non–enteric coated) should be chewed and swallowed immediately
β-Blockade	• Decreases risk of ventricular arrhythmias and sudden cardiac death in early post-MI period • Decreases HR and myocardial oxygen demand and increases diastolic filling time of ventricles, thereby improving oxygen flow through the coronary arteries **β-Blockade should be initiated within the first 24 hr of an ACS <u>unless:</u>** • There are signs of HF • Active evidence of other shock states • If at increased risk of cardiogenic shock (SBP < 120 mm Hg, HR > 110 beats/min or < 60 beats/min, age > 70, and increased time since onset of symptoms) ***Relative contraindications* to β-blockade include:** • PR interval > 0.24 s, second- or third-degree heart block, and active asthma/reactive airway disease **Note: Intravenous β-blockers may be particularly harmful in patients with risk factors for shock.*
Helpful acronym MONA +/- BB	• Morphine • Oxygen • Nitrates (unless contraindicated) • Aspirin • β-Blocker (unless contraindicated)
Other precautions	• ACE inhibitors/ARBs should be used with caution within the first 24 hr of ACS because of the risk of hypotension and potential contribution to contrast-induced nephrotoxicity • Any NSAID other than aspirin should be avoided and/or discontinued for reasons beyond GI bleeding and nephrotoxicity, which may include reinfarction, hypertension, HF exacerbation, myocardial rupture, and overall increased risk of mortality associated with their use

ACS = acute coronary syndrome(s); GI = gastrointestinal; HF = heart failure; IV = intravenous; LMNOP = lasix-morphine-nitro-oxygen-position/positive pressure ventilation; NSAID = nonsteroidal anti-inflammatory drug; RV = right ventricular; Sao_2 = arterial oxygen saturation.

D. Revascularization
   1. Nonsurgical – Details of interventional cardiology procedures are too broad to be discussed in great detail in this chapter; however, the following are some considerations of the intervention that may play a role in post-procedural management during a left heart catheterization and percutaneous coronary intervention (PCI):
      a. Access site
         i. Radial artery (and, rarely, brachial artery) (Catheter Cardiovasc Interv 2011;78:840-6)
            (a) Easily accessible
            (b) Increased risk of vasospasm during procedure
            (c) Easily compressible vessel when hemostasis is needed post-procedure
            (d) Does not prevent patient mobility post-procedure
         ii. Femoral artery
            (a) Easily accessible
            (b) More difficult to compress vessel when hemostasis is needed post-procedure and also associated with increased bleeding complications

(c) Limits patient mobility post-procedure for at least 12–24 hours (bleeding risk after sheath removal)

b. Common interventions performed:
   i. Stent placement
      (a) Important to note the number of stents placed, types of stents, and locations of placement
      (b) Bare metal stent
         (1) Requires aspirin for life and a P2Y12 antagonist for at least 1 month (12 months preferred) to allow adequate time for endothelialization of the stent(s)
         (2) Longer therapies may be considered, depending on the number of stents and location(s) of the stent(s).
         (3) Higher risk of in-stent stenosis over time (because of neointimal cell proliferation)
      (c) Drug-eluting stent
         (1) Requires aspirin for life and a $P2Y_{12}$ antagonist for at least 6 months (12 months preferred). Longer or shorter therapy durations may be considered, depending on the number and/or location(s) of the stent(s) as well as patient bleeding risk. Some patients may benefit from a modified DAPT regimen, such as discontinuing aspirin after 1–3 months of DAPT and continuing with $P2Y_{12}$ inhibitor monotherapy for the remaining 9–11 months (Circulation 2022;145:e18-114). Of course, an individualized approach to DAPT duration is necessary.
         (2) The benefit of drug-eluting stents (i.e., everolimus, zotarolimus, biolimus) is the mitigation of in-stent restenosis, which is more common with bare metal stenting. However, the rate of stent endothelialization is also impaired such that the risk of stent thrombosis persists for a longer period. Consequently, drug-eluting stents mandate longer-term dual antiplatelet therapy than bare metal stents.
   ii. Thrombectomy: Thrombus aspiration generally followed by stent placement at site of lesion
   iii. Percutaneous transluminal coronary angioplasty (PTCA), also called "plain old balloon angioplasty" (POBA): Balloon expansion and at least temporary displacement of occlusion at site of lesion

2. Surgical – Details of coronary artery bypass grafting (CABG) procedures, including conduit type and use of cardiopulmonary bypass (or performing off pump), are too broad to be discussed in greater detail in this chapter. According to the Society of Thoracic Surgeons/American College of Cardiology/American Heart Association (STS/ACC/AHA) CABG guidelines, emergency CABG is recommended in patients with an acute MI in the following scenarios (J Am Coll Cardiol 2011;58:e123-210):
   a. Primary PCI has failed or cannot be performed
   b. Coronary anatomy is more suitable for CABG
   c. Persistent ischemia of a significant area of myocardium at rest and/or hemodynamic instability refractory to nonsurgical therapy is present
   d. Requirement of surgical repair of a postinfarction mechanical complication of MI (i.e., ventricular septal rupture, mitral valve insufficiency caused by papillary muscle infarction and/or rupture, or free wall rupture)
   e. Patients with cardiogenic shock who are suitable for CABG, irrespective of the time interval from MI to onset of shock and time from MI to CABG
   f. Patients with life-threatening ischemic ventricular arrhythmias in the presence of a left main stenosis of 50% or more and/or three-vessel CAD

g. CABG use is reasonable as a revascularization strategy in patients with multivessel CAD with recurrent angina or MI within the first 48 hours of STEMI presentation as an alternative to a more delayed strategy.
h. Early revascularization with PCI or CABG is reasonable for select patients older than 75 years with ST-segment elevation or left bundle branch block who are suitable for revascularization irrespective of the time interval from MI to onset of shock.

3. Medical management – No revascularization:
   a. Less invasive strategies may be opted for rather than revascularization in some patients. Considerations reinforce a patient-centered approach driven by evidence of ongoing/recurrent ischemia and feasibility of revascularization, including myocardial viability, patient frailty, and comorbid disease states.
   b. Ongoing care outlined in section E remains the focus of optimizing aggressive medical management.

E. Antithrombotics in MI
   1. The roles and combinations of antithrombotics continue to be refined in select populations (those with NSTEMI/ACS, STEMI, and PCI). For the most current guidelines and landmark trials, please see www.acc.org/guidelines.
   2. Oral antiplatelet therapy
      a. Platelets can be activated by several different mechanisms, only some of which can be inhibited by medications.
      b. Evaluation of antiplatelet therapy can be performed with platelet function testing and/or genotyping. However, neither are currently performed routinely in the clinical setting. Clinical outcomes with the use of platelet function testing to modify antiplatelet therapy (i.e., high-dose vs. standard-dose clopidogrel) in PCI patients with high on-treatment platelet reactivity have been negative to date (JAMA 2011;305:1097-1105). Outcomes with prospective genotype-guided antiplatelet therapy (CYP2C19) from large cohorts of PCI patients have had lower rates of major adverse cardiovascular events and lower rates of bleeding (JACC Cardiovasc Interv 2018;2:181-91; J Am Coll Cardiol 2018;71:1869-77; N Engl J Med 2019;381:1621-31).
      c. The response to antiplatelets in certain critical care patient scenarios, such as acute hepatic and kidney injury, have not been well documented. In other situations, such as when enteral administration is not an option, parenteral antiplatelets may be considered (Table 5). These options include glycoprotein IIb/IIIa (GP IIb/IIIa) inhibitors and cangrelor, which have been studied primarily in the setting of PCI but have data supporting their use as bridging agents.

Table 4. Oral Antiplatelet Therapy (Circulation 2013;127:e362-425; Circulation 2014;130:e344-426; J Am Coll Cardiol 2011;58:e123-210; J Am Coll Cardiol 2011;58:e44-122; Br J Clin Pharmacol 2011;72:647-57)

Medication	Aspirin	Clopidogrel	Prasugrel	Ticagrelor
Mechanism of action	Inhibits thromboxane A_2–mediated platelet activation	Inhibits ADP-mediated platelet activation at $P2Y_{12}$ receptor	Inhibits ADP-mediated platelet activation at $P2Y_{12}$ receptor	Inhibits ADP-mediated platelet activation at $P2Y_{12}$ receptor
Loading dose	162–325 mg	300-600 mg	60 mg	180 mg
Maintenance dose	81 mg daily	75 mg daily	5–10 mg daily[a]	90 mg BID
Route	Oral	Oral	Oral	Oral
Prodrug	No	Yes	Yes	No
Reversible platelet binding	No	No	No	Yes
Onset	30 min	2–6 hr	30 min	30 min
% Platelet inhibition	~ 10–20	30–40	60–70	60–70
Recommended holding duration before CABG[b]	Do not hold	5 days	7 days	3 days
Other notable adverse effect(s) or clinical pearls		Pharmacogenomic variability (CYP2C19) in response is well documented	Contraindication in patients with any history of stroke or TIA because of bleeding risk, and warning for use in patients ≥ 75 years old or weight < 60 kg	• Adenosine-induced dyspnea and bradyarrhythmias • Older adult patients and patients with moderate or severe hepatic impairment may be at increased risk of bleeding • Concomitant maintenance doses of aspirin > 100 mg should be avoided because of lack of efficacy

[a]Patients with lean body weight (< 60 kg) should receive a 5 mg daily maintenance dose of prasugrel

[b]Decisions to hold these agents before other invasive procedures must consider indications for use, risk of thrombosis (i.e., stent type, locations, and number), and risk of bleeding associated with intended procedure/surgery (JAMA 2013;310:1451-2; Catheter Cardiovasc Interv 2013;82:1113-4; JAMA 2013;310:189-98; N Engl J Med 2013;368:2113-24; J Am Coll Cardiol 2011;58:e44-122; Reg Anesth Pain Med 2010;35:64-101; J Am Coll Cardiol 2007;49:734-9).

ADP = adenosine diphosphate; BID = twice daily; TIA = transient ischemic attack.

3. Parenteral antithrombotics
    a. Use of these agents is most concentrated in the procedural setting, although use may continue for a finite period post-procedurally.
    b. The selection and use among the agents in Table 5 may depend on presentation, timing/dose of pre-procedural antiplatelet medication administration, clot burden during procedure, and estimated bleeding risk of the procedure.

Table 5. Parenteral Antithrombotics (Circulation 2014;130:e344-426; Circulation 2013;127:e362-425;Circulation 2015;131:1123-49; J Am Coll Cardiol 2011;58:e123-210; J Am Coll Cardiol 2011;58:e44-122; Br J Clin Pharmacol 2011;72:647-57)

Medication	Heparin (UFH)[a]	Bivalirudin	Eptifibatide	Tirofiban	Cangrelor[b]
Mechanism of action	Indirect thrombin inhibition (binds to antithrombin III)	Direct thrombin inhibitor	Binds to GP IIb/IIIa platelet receptor, inhibiting final common pathway for platelet aggregation	Binds to GP IIb/IIIa platelet receptor, inhibiting final common pathway for platelet aggregation	Inhibits ADP-mediated platelet activation at $P2Y_{12}$ receptor
Bolus dose	50–100 units/kg based on target activated clotting time (ACT)	0.75 mg/kg	180 mcg/kg x 2, 10 min apart	25 mcg/kg	30 mcg/kg
Continuous infusion	Not typically given during PCI	1.75 mg/kg/hr	2 mcg/kg/min	0.15 mcg/kg/min	4 mcg/kg/min
Platelet inhibition	Indirectly inhibits thrombin-mediated platelet activation	Indirectly inhibits thrombin-mediated platelet activation	Restoration of platelet function within 6–8 hr of discontinuation	Restoration of platelet function within 6–8 hr of discontinuation	Onset of platelet inhibition within min of administration Restoration of platelet function within 1 hr of discontinuation
Elimination	Hepatic and reticulo-endothelial system	80% plasma proteolysis 20% renal	Renal (50%)	Renal (65%)	Plasma dephosphorylation
Removed by dialysis	No	Partial	Yes	Yes	N/A
Other notable adverse effect(s) or clinical pearls	Target ACT of about 200–300 s during procedure based on presence of GP IIb/IIIa inhibitor and device used	Requires renal dose adjustment Useful in patients with suspected or confirmed HIT May be preferred in patients with higher bleeding risk (particularly compared with UFH with GP IIb/IIIa inhibitor and among patients who undergo a transfemoral PCI approach)	Requires renal dose adjustment Acute, profound thrombocytopenia	Requires renal dose adjustment Acute, profound thrombocytopenia	Oral prasugrel and clopidogrel must be given immediately after discontinuation of cangrelor infusion to maintain platelet inhibition. Ticagrelor should be given as soon as possible, and administration may overlap with cangrelor infusion to maintain platelet inhibition after cessation of cangrelor

[a]Enoxaparin may be used as an alternative to UFH.

[b]U.S. Food and Drug Administration (FDA) approved as an adjunct to PCI for reducing the risk of periprocedural myocardial infarction (MI), repeat coronary revascularization, and stent thrombosis (ST) in patients who have not been treated with a $P2Y_{12}$ platelet inhibitor and are not receiving a glycoprotein IIb/IIIa inhibitor.

ACT = activated clotting time; GP = glycoprotein; HIT = heparin-induced thrombocytopenia; N/A = not applicable; UFH = unfractionated heparin.

F. Post-Intervention Complications
   1. Bleeding (particularly retroperitoneal bleeding)
      a. Several antithrombotic agents are used during PCI to inhibit both the platelets and the clotting cascade, causing potential coagulopathies.
      b. In addition to antithrombotic use, the catheterization access site has been identified as a major contributor to post-PCI bleeding complications. Note that use of the trans-radial approach has been demonstrated to be a bleeding avoidance strategy compared with the trans-femoral approach.
   2. Dissection/rupture of free wall, coronary artery, or aorta: Although this may be spontaneous, it may also be caused by vessel trauma from the catheter itself.
   3. Stent thrombosis
      a. When antiplatelet therapy is discontinued early (aspirin, $P2Y_{12}$ inhibitor, or both), stent thrombosis may occur in up to 25% of coronary artery stents, irrespective of type of stent (drug-eluting stent or bare metal stent).
      b. Almost 1 in 7 patients may discontinue $P2Y_{12}$ inhibitors within 30 days post-PCI, thus increasing mortality risk (adjusted hazard ratio [HR] 9.0; 95% confidence interval [CI], 1.3–60.6) (JAMA 2013;310:189-98).
      c. Mortality rates associated with stent thrombosis can be as high as 45%.
      d. Despite bleeding risks in critically ill patients, careful consideration should be given to correlating these risks with the risk of stent thrombosis.
   4. Papillary muscle rupture and mitral regurgitation (mechanical complications)
   5. Arrhythmias (particularly after reperfusion)
   6. Contrast-induced nephropathy

G. Ongoing Care – Quality measures for NSTEMI or STEMI independent of revascularization or medical management
   1. Medications that should be initiated before discharge or contraindications should be documented in the medical record:
      a. Aspirin
      b. Statin (high intensity)
      c. $P2Y_{12}$ inhibitor
      d. β-Blocker
      e. If LVEF less than or equal to 40%
         i. ACE inhibitor or ARB or angiotensin receptor/neprilysin inhibitor
         ii. Aldosterone antagonist (if also evidence of HF and/or diabetes)
         iii. Sodium glucose co-transporter 2 inhibitor (if patient has other compelling indications such as diabetes or chronic kidney disease)
   2. Interventions and/or referrals
      a. LV function assessment (by imaging or during catheterization)
      b. Cardiac rehabilitation
      c. Smoking cessation counseling
      d. Measurement of a lipid profile, including the low-density lipoprotein (LDL) cholesterol, should preferably be obtained within 24 hours of admission. Any lipid profile measured between 6 months before first medical contact and hospital discharge qualifies for this quality measure.
   3. For more information on cardiology-related quality measures and registries, see www.ncdr.com.

Patient Case *(Continued)*

4. J.M.'s ECG results reveal no acute evidence of ST segment changes. However, the resident is still considering a diagnosis of ACS, given the patient's shortness of breath and mild troponin elevation. Which statement would be most accurate regarding the potential for other potential diagnoses?
   A. No, this is most likely an NSTEMI.
   B. Yes, it is likely undiagnosed chronic obstructive pulmonary disease.
   C. Yes, it is likely early sepsis.
   D. Yes, it is likely decompensated HF.

5. J.M. has been experiencing intermittent bradycardia on telemetry. The team has consulted the cardiac electrophysiology team. In the meantime, which statement most accurately reflects whether any other underlying correctable/contributing causes can be addressed?
   A. Ticagrelor could be discontinued and switched back to clopidogrel.
   B. Ticagrelor should be discontinued altogether.
   C. No treatment intervention is needed; this is likely the result of J.M.'s HF progression.
   D. J.M.'s hyperkalemia should be treated.

V. ARRHYTHMIAS AND ANTIARRHYTHMICS

A. Pathophysiology: Arrhythmias are generally caused by automaticity and/or re-entrant conduction abnormalities.

B. Bradyarrhythmias (beyond sinus) and Types of Heart Block (Critical Care Medicine 2014:xix; Pathophysiology of Heart Disease 2011:xiv; BMJ 2002;324:662-5; BMJ 2002;324:415-8)
   1. Etiologies of heart block (Box 3)

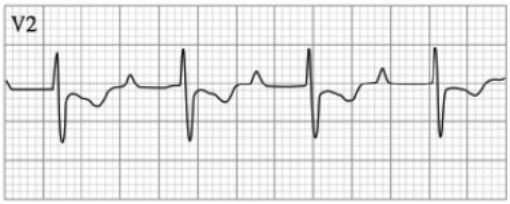
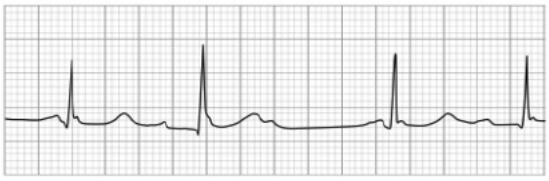
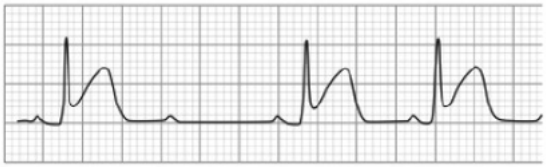
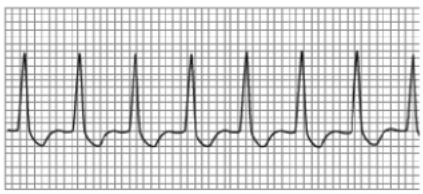
Box 3. Causes of Bradyarrhythmias

- CAD
- Degenerative conduction disease
- Drug induced
- Electrolyte disturbances (particularly hyperkalemia)
- Endocarditis
- Myocarditis
- Surgery (particularly cardiac surgery)
- Tumors
- Vagus nerve–mediated heart block

CAD = coronary artery disease.

   2. When evaluating ECGs in the presence of heart block, QRS complex evaluation can guide some differential diagnoses to the source of conduction problems (see Table 6).
      a. Narrow QRS complexes commonly indicate AV nodal dysfunction.
      b. Wide QRS complexes may indicate dysfunction in either the AV node or the His-Purkinje system.

Table 6. Bradyarrhythmias and Types of Heart Block (Critical Care Medicine 2014:xix; Pathophysiology of Heart Disease 2011:xiv; BMJ 2002;324:662-5; BMJ 2002;324:415-8)

Type	ECG Example	Description
First degree		Delayed conduction from the sinoatrial (SA) node to the atrioventricular (AV) node characterized by a P-R interval > 0.2 s Relatively benign; however, underlying contributors should be evaluated and minimized (i.e., β-blockers and other AV nodal blocking agents)
Second-degree Mobitz type 1 (Wenckebach)		Consistent P-P interval with progressive prolongation of the P-R (indicating impaired SA to AV node conduction) eventually resulting in absence of a QRS complex because of the lack of AV node conduction of atrial impulse Of most concern in older adult patients in whom this may be indicative of progressive conduction disease; may be more benign in younger patients "Longer, longer, longer, drop … must be Wenckebach"
Second degree Mobitz type 2		Consistent P-P interval and consistent P-R interval duration with spontaneous absence of a QRS complex because of the lack of AV node conduction of atrial impulse Usually indicative of more significant conduction disease and associated with syncope, HF, and increased mortality rates
Third degree (Complete heart block)		Characterized by consistent P-P intervals, consistent R-R intervals, and variable/ random P-R interval representing independent, uncoordinated atrial and ventricular conduction (A-V dissociation). The arrows on this ECG strip identify P waves, showing regular P-P interval with varying P-R interval
Junctional rhythm		Manifested when sinus node dysfunction allows the AV node to take over as the active cardiac pacemaker, resulting in retrograde conduction through the atria

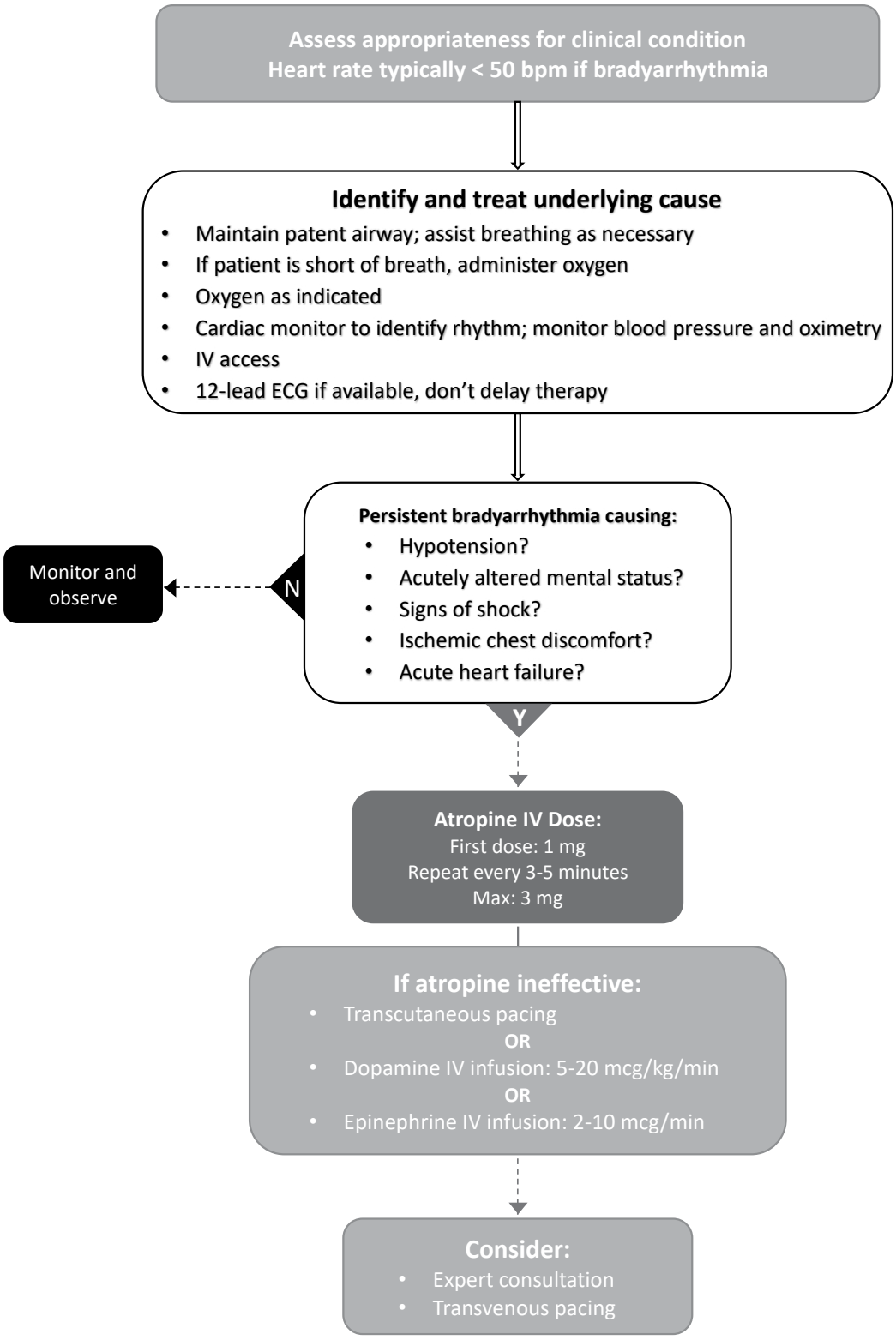


Figure 8. Advanced cardiac life support bradycardia algorithm adapted from the 2020 American Heart Association guidelines for Cardiopulmonary Resuscitation and Emergency Cardiovascular Care (Circulation 2021;142(14_suppl_2):S366-468).

3. Management of bradyarrhythmias includes three principal strategies (see also Figure 8). Stabilize the patient, if symptomatic.
   a. Consider atropine for temporary correction to decrease vagal tone. Note that the recommended dose of atropine is 1 mg, which can be given every 3–5 minutes for a maximum of 3 mg.
   b. Consider pacing strategies (temporary transvenous pacer, transcutaneous pacer, pacing pulmonary artery catheter). Interrogate permanent pacemaker for malfunction/optimization.
   c. Consider chronotropic β-agonist infusion.
      i. Dopamine
      ii. Epinephrine
      iii. Isoproterenol
      iv. Dobutamine (if blood pressure normal or elevated)
   d. Identify and treat underlying causes/toxidromes.

C. Tachyarrhythmias (beyond sinus) – Etiologies and hemodynamic consequences
   1. Etiologies of tachyarrhythmias
      a. Usually related to enhanced automaticity, reentry, or triggered activity
      b. A history that includes ischemic heart disease or congestive cardiac failure is 90% predictive of VT.
   2. In evaluating ECGs for tachyarrhythmias, some fundamental considerations include:
      a. Evaluate for the presence of P waves.
      b. Evaluate the width of the QRS complex.

Table 7. Tachyarrhythmias (beyond sinus)[a] (Circulation 2014;130:2071-104; BMJ 2002;324:594-7776-9; BMJ 2002;324:1264-7; BMJ 2002;324:1201-4; Circulation 2012;125:381-9; Heart Fail Rev 2014;19:285-93; J Am Coll Cardiol 2018;72:1677-749)

Supraventricular Tachyarrhythmias				
Type	**Rhythm**	**P-wave Attributes**	**Atrial Rate (beats/min)**	**Description**
Premature atrial complexes (PACs)	Irregular	N/A	N/A	Generally benign but may be more evident with increased sympathetic tone, stress, and pericarditis or with sympathomimetic use In some cases, can lead to an AV block or initiate a reentrant supraventricular tachycardia (SVT) or AF
Supraventricular tachycardia (SVT)	Regular	Hidden or can be retrograde	140–250	Usually sudden onset/offset with narrow QRS complexes Often caused by reentry within the atrium or AV node or by an accessory conduction pathway Can be subcategorized as: AV nodal reentrant tachycardia (AVNRT) AV reentrant tachycardia (AVRT) Sinus node reentry tachycardia
Atrial flutter (AFl)	Regular	Saw-tooth appearance	180–350	Generally conducts through the ventricles in a 2:1 fashion, resulting in ventricular rates of 100–150 beats/min In some scenarios, slowing the atrial rate may increase the number of conducted beats, leading to rapid ventricular rates and potential hemodynamic compromise Associated with increased risk of stroke

Table 7. Tachyarrhythmias (beyond sinus)[a] (Circulation 2014;130:2071-104; BMJ 2002;324:594-7776-9; BMJ 2002;324:1264-7; BMJ 2002;324:1201-4; Circulation 2012;125:381-9; Heart Fail Rev 2014;19:285-93; J Am Coll Cardiol 2018;72:1677-749) *(continued)*

Supraventricular Tachyarrhythmias				
Type	**Rhythm**	**P-wave Attributes**	**Atrial Rate (beats/min)**	**Description**
Atrial fibrillation (AF)	Irregular	No distinct P wave visible	Unable to determine	Most common arrhythmia, characterized by irregular ECG appearance because of multiple reentry circuits and ectopic foci ("irregularly irregular") Often associated with structural heart disease and potentiated by increased left atrial pressures among other influencing contributors such as age, inflammation, and sympathetic tone Associated with increased risk of stroke
Multifocal atrial tachycardias (MATs)	Irregular	> 3 different types of distinct P waves	100–130	Can be misdiagnosed as AF Commonly associated with respiratory disease, HF, critical illness May be exacerbated by electrolyte abnormalities or toxicity with digoxin or theophylline
Ventricular Arrhythmias				
Type				**Description**
Premature ventricular complexes (PVCs)				Results from an ectopic ventricular focus conduction that can be identified by the lack of a preceding P wave. Commonly benign or asymptomatic but may be of concern if present in patterns or in patients with advanced heart disease Bigeminy: Every other beat is a PVC Trigeminy: Every third beat is a PVC Couplets: Patterns of two consecutive PVCs Triplets: Patterns of three consecutive PVCs
Ventricular tachycardia (VT)				Potentially lethal wide QRS complex tachycardia characterized according to morphology and duration; can degenerate into ventricular fibrillation (VF) or asystole *Monomorphic VT:* When every QRS complex appears the same and the rate is regular (100–200 beats/min); commonly caused by reentry circuit related to myocardial scar or fibrosis *Polymorphic:* When the QRS complexes continually vary in shape and rate; may be related to multiple ectopic foci, myocardial ischemia, or Torsades de pointes from prolongation of QT interval. *Non-sustained VT:* Self-terminating episodes lasting for < 30 s *Sustained VT:* If VT persists for more than 30 s, produces severe symptoms, including syncope, or requires termination by administration of an antiarrhythmic drug or direct cardioversion/defibrillation
Ventricular fibrillation (VF)				Life-threatening arrhythmia with a chaotic ECG with no discernible QRS, representing rapid disorganized conduction with no resultant coordinated ventricular contractions

[a]Other arrhythmias beyond the scope of this review include sick sinus syndrome (tachy-brady syndrome) and Wolff-Parkinson-White.

3. Management of tachyarrhythmias with a pulse includes the following strategies:
   a. Stabilization of patient: See Figure 9
      i. Atrial tachyarrhythmias
         (a) Heart rate control typically with class II agents, class IV agents, and/or digoxin (see Appendix A)
         (b) Heart rhythm control
            (1) Antiarrhythmic therapy typically with class Ic or class III agents (see Appendix A)
            (2) Synchronized electrical cardioversion
            (3) Catheter-based ablation
            (4) Surgical ablation
      ii. Ventricular tachyarrhythmias
         (a) Catheter ablation
         (b) Antiarrhythmic therapy
         (c) Evaluation for implantable cardioverter-defibrillator
   b. Overall considerations for antiarrhythmics (see Appendix A) (Circulation 2014;130:2071-104; Heart Fail Rev 2014;19:285-93; Circulation 2012;125:381-9; BMJ 2002;324:594-7; BMJ 2002;324:776-9; BMJ 2002;324:1264-7; BMJ 2002;324:1201-4)
      i. Abrupt discontinuation of chronic antiarrhythmics (i.e., chronic medication before ICU admission) should be done with caution and awareness of antiarrhythmic indication as well as risks-benefits of continuation/cessation.
      ii. Appropriate monitoring and potential adjustment may be warranted with many of these agents in the critically ill patient.
         (a) QT/QTc prolongation increases the risk of Torsades de pointes. This risk can be influenced not only by the absolute duration of the QT/QTc but also by the rate at which the QT/QTc is changed (i.e., faster change may also increase risk).
         (b) The class III antiarrhythmics sotalol and dofetilide can be used for a variety of arrhythmias (most notably AF/AFl) and, because of some notable characteristics, should be watched closely when used in the critically ill patient.
            (1) Both are renally eliminated.
            (2) Both have notable interactions with several other medications (especially dofetilide on the basis of CYP3A4 and renal transport) in addition to their effect on the QT/QTc interval.
            (3) ECG, electrolytes, and renal function monitoring should be performed for safety and tolerability 2–3 hours after the first five doses when initiating, reinitiating, or introducing another interacting or QT-prolonging agent.
            (4) Electrolyte abnormalities may place a patient at increased risk of Torsades de pointes (particularly magnesium and potassium).
            (5) Dofetilide is particularly useful among patients with structural heart disease (including HF) because of a neutral effect on mortality. In contrast, sotalol has been associated with increased mortality in patients with HF and is not recommended in this patient population for atrial arrhythmias.
         (c) If a patient develops Torsades de pointes, the patient should be managed with the following:
            (1) Intravenous magnesium 2-mg rapid infusion
            (2) Discontinuation of any offending medications contributing to QT prolongtion
            (3) Pharmacologic pacing with beta agonists (i.e., isoproterenol, dobutamine, epinephrine, dopamine); bradycardia increases the risk for TdP, so increasing HR can help prevent future events
            (4) Correction of electrolyte disturbances such as hypokalemia
            (5) Temporary pacing to increase heart rate

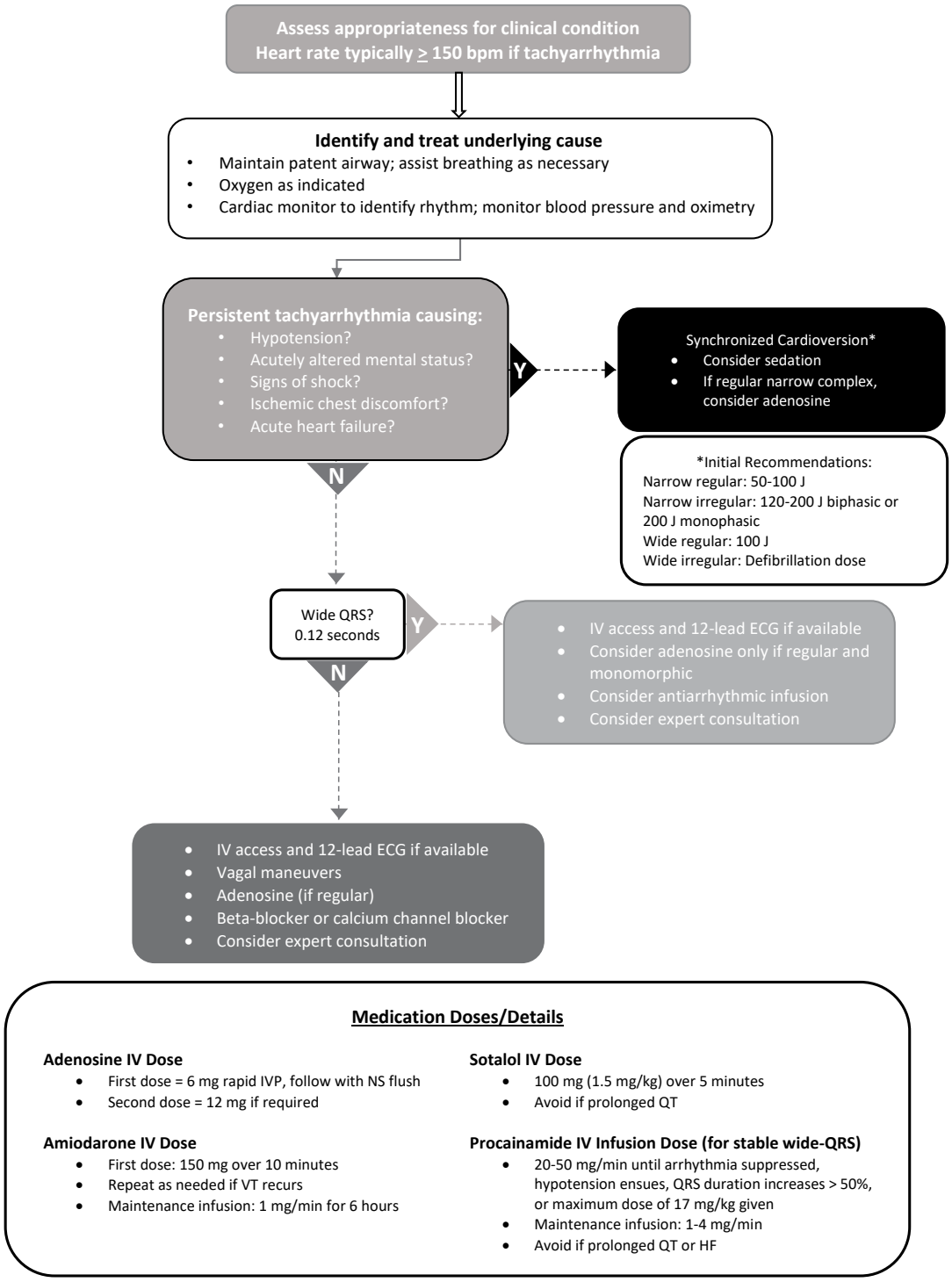


Figure 9. Advanced cardiac life support tachycardia algorithm adapted from the 2020 American Heart Association guidelines for Cardiopulmonary Resuscitation and Emergency Cardiovascular Care (Circulation 2021;142(16_suppl_2):S366-468).

iii. Common agents in the ICU used to treat tachyarrhythmias
  (a) β-Blockers
    (1) May be of limited use in patients receiving vasopressors and/or inotropes, but can be used for both atrial and ventricular tachyarrhythmias
    (2) Esmolol has the shortest half-life (cleared by plasma esterases), and administration rates commonly coincide with high volumes of fluid.
    (3) Agents with combined α_1-antagonism and nonselective β-antagonism will have greater effects on blood pressure than their β_1-specific antagonist counterparts.
      - β_1-specific antagonist examples:
        - Esmolol (intravenous), metoprolol tartrate (oral/intravenous)
      - Agents with combined α_1- and nonselective β-antagonism effects include:
        - Carvedilol (oral) – about 25:1 ratio of β:α_1 receptor activity
        - Labetalol (intravenous) – about 7:1 ratio of β:α_1 receptor activity
        - Labetalol (oral) – about 3:1 ratio of β:α_1 receptor activity
  (b) Non-dihydropyridine calcium channel blockers (diltiazem, verapamil)
    (1) Not recommended in patients with HF with reduced ejection fraction (HF*r*EF) (systolic HF) because of potent negative inotropic effects (class III C). (J Am Coll Cardiol 2013;62:e147-239)
    (2) May be of limited use in patients on vasopressors and/or inotropes, but can be used for atrial tachyarrhythmias
  (c) Amiodarone
    (1) Commonly used for both atrial and ventricular tachyarrhythmias
    (2) Demonstrated safety in patients with structural heart disease (CHF) because of neutral effect on mortality
    (3) Affects all phases of the action potential (sodium [Na], calcium [Ca], potassium [K] channel and provides some α- and β-antagonism)
    (4) Severe hypotension can occur with intravenous amiodarone because of the solvent polysorbate 80 (Tween 80) used in some formulations. This transient hypotension in patients with a pulse can be associated with rapid infusions, particularly of undiluted drugs, and is best avoided by dilution and slower administration. However, in pulseless patients, rapid intravenous administration of undiluted amiodarone is commonly indicated.
    (5) Among common agents used in the ICU, amiodarone has one of the largest volumes of distribution (about 60 L/kg) and prolonged half-lives (about 35–110 days).
    (6) In adults, common total intravenous or oral loading doses throughout several days are about 8–10 g before switching to a maintenance dose.

$$\text{Loading dose} = \text{concentration} \times \text{volume of distribution} \times \text{weight}$$
$$7.2\text{–}12\ \text{g} = (1.5\text{–}2.5\ \text{mg/L}) \times (60\ \text{L/kg}) \times (80\ \text{kg})$$

    (7) Extensive metabolism by CYP3A4 and CYP2C8 and enzyme inhibitor of CYP3A4, CYP1A2, CYP2C9, and CYP2D6, resulting in several drug-drug interactions ((e.g., warfarin, statins)
    (8) If patients are to be continued on amiodarone for a prolonged duration, thyroid and liver function tests, pulmonary function tests, chest radiographs, and ophthalmic examinations should be evaluated periodically (Heart Rhythm 2007;4:1250-9).
  (d) Lidocaine
    (1) Indicated only for treatment of ventricular tachyarrhythmias and is particularly useful if the tachyarrhythmia is caused by active ischemic myocardial tissue

(2) Efficacy and toxicity are both concentration dependent.
(3) Metabolism is largely dependent on hepatic blood flow; the primary metabolites are monoethylglycinexylidide (MEGX) and glycine xylidide (GX) and are mediated by CYP1A2. Both lidocaine and MEGX contribute to therapeutic effect and toxicity, whereas GX has predominantly toxic adverse effects. Lidocaine's metabolites are eliminated renally.
(4) Concentration-dependent protein binding includes about 25% bound to albumin and about 50% bound to α_1-acid glycoprotein (AAG).
(5) As AAG increases and decreases as an acute-phase reactant within the first 12–72 hours after certain stresses (e.g., acute MI, HF exacerbation, trauma), an unsuspected variation can occur in free lidocaine concentrations. Because of the toxicity profile and concentration-dependent efficacy, routine concentrations should be monitored in most patients if lidocaine is to be continued beyond 24 hours (particularly in patients with advanced age, HF, liver disease, and renal dysfunction).

Table 8. Lidocaine Concentrations and Toxicity Symptoms

Total Serum Concentration (mg/L)	Toxicity
< 1.5–5	Typical therapeutic goal concentration
> 5–8	Increasing risk of light-headedness, confusion, dizziness, tinnitus, twitching, tremor, blurred or double vision, hypotension, and bradycardia
> 8	High risk of seizures, respiratory depression, hypotension, bradycardia, AV nodal blockade, decreased cardiac output

AV = atrioventricular.

(e) Procainamide
(1) Indicated for the treatment of hemodyanmically stable, sustained, monomorphic VT and atrial fibrillation with pre-excitation (ESC VT and AF guidelines)
(2) Compared to amiodarone, procainamide has been shown to be more effective at terminating monomorphic VT with fewer cardiac complications (Eur Heart J 2017;38:1329-35).
(3) Procainamide should be avoided in patients with severe heart failure, acute myocardial infarction, and end-stage renal disease
(4) Procainamide is a sodium channel blocking antiarrhythmic, so close monitoring because of the QRS complex is necessary during treatment. Procainamide should be discontinued if the QRS widens by more than 50%.
(5) Close monitoring of the QT/QTc is also important because both procainamide and its active metabolite, N-acetyl procanamide, have potassium channel blocking properties.
(f) Digoxin
(1) Digoxin may be considered to slow a rapid ventricular response in patients with ACS and AF associated with severe LV dysfunction and HF or hemodynamic instability (Circulation 2019;140:e125-51).
(2) Use for AF should be done with caution. Two published retrospective trials have shown considerable evidence for increased hospitalization rates and risk of death.
- 71% increased risk of death (HR 1.71; 95% CI, 1.52–1.93) and 63% increased risk of hospitalization (HR 1.63, 95% CI, 1.56–1.71) (Circ Arrhythm Electrophysiol 2015;8:49-58)

- Digoxin-treated patients had higher mortality rates (95 vs. 67 per 1000 person-years; p<0.001), and use was independently associated with mortality, despite multivariate adjustment (HR 1.26; 95% CI, 1.23–1.29, p<0.001) (J Am Coll Cardiol 2014;64:660-8).
- Eliminated renally; thus, may pose risks of digoxin toxicity in patients with acute or chronic renal failure

(3) Although the volume of distribution is relatively large (7–10 L/kg in healthy adults), only a small percentage of total body stores are present in the serum.
(4) Digoxin has a smaller volume of distribution in patients with renal failure (around 4.5 L/kg); thus, loading doses may be lower in these patients.
(5) Digoxin efficacy and toxicity do not correlate well with drug concentrations.
- Common therapeutic targets for heart rate control are 0.8–1.5 ng/mL, although many clinical laboratories report therapeutic concentrations within 0.5–2.0 ng/mL. For patients with HFrEF, the recommended serum concentration is 0.5−0.9 ng/mL
- Toxicity, which can present at any serum concentration, should be evaluated according to clinical manifestations, including: nausea, vomiting, anorexia, mental status changes, visual disturbances, ventricular arrhythmias, bradycardia, and hyperkalemia.
- Efficacy is largely based on clinical control of the heart rate; steady-state concentrations (more than 5–7 days after initiation) are typically used only to validate that concentrations are not supratherapeutic.

4. Anticoagulation for AF or atrial flutter
   a. According to the 2023 ACC/AHA/ACCP/HRS Guideine for the Diagnosis and Management of Atrial Fibrillation Guidelines all patients with atrial flutter or paroxysmal, persistent, or permanent AF should be evaluated for anticoagulation, preferably using the CHA_2DS_2-VASc score to approximate stroke risk (Circulation 2024;149:e1-156).
      i. For a CHA_2DS_2-VASC score of 2 or greater in men and 3 or greater in women, an oral anticoagulant is recommended (J Am Coll Cardiol 2019;12:104-32).
      ii. It is reasonable to omit anticoagulation in patients with a CHA_2DS_2-VASc score of 0 in men or 1 in women. Antiplatelet therapy is not recommended for stroke prevention in atrial fibrillation patients, even in those who are low risk.
      iii. For patients with AF and a mechanical prosthetic heart valve or valves, warfarin anticoagulation and international normalized ratio (INR) goals should be consistent with the type and location of the prosthetic valve. Bridging is no longer routinely recommended (Chest 2022;165:1127-39). Bridging with unfractionated heparin can be considered in those at high risk for thromboembolic events, including:
         (a) Older generation mechanical valves (e.g., tilting-disc valve)
         (b) A mechanical mitral valve with one or more other risk factors for thromboembolism
         (c) A recent thromboembolic event in preceding 3 months
         (d) History of a perioperative thromboembolism
      iv. In patients with AF at intermediate risk for thromboembolism (average $CHADS_2$ score of 2-3), the use of bridging with low-molecular-weight heparin was non-inferior to no bridging for elective procedures. However, there was a significant excess of major bleeding in the low-molecular-weight heparin group. Note that patients with a mechanical valve or stroke/transient ischemic attack/systemic embolism within the preceding 12 weeks were not eligible for inclusion in the study (N Engl J Med 2015;373:823-33).

v. Although the $CHADS_2$ and CHA_2DS_2-VASc scores are commonly used to estimate annual stroke risk, these scoring systems were not founded in the context of critically ill patients.

Table 9A. $CHADS_2$ Stroke Risk Score in AF

Risk Assessment	Score	Total Patient Score	Adjusted Annual Stroke Rate, %
$CHADS_2$			
Congestive heart failure	1	0	1.9
Hypertension	1	1	2.8
Age ≥ 75	1	2	4.0
Diabetes	1	3	5.9
Stroke/TIA/thromboembolism	2	4	8.5
Maximum score	6	5	12.5
		6	18.2

Table 9B. CHA_2DS_2-VASc Stroke Risk Score in AF

Risk Assessment	Score	Total Patient Score	Adjusted Annual Stroke Rate, %
CHA_2DS_2-VASc			
Congestive heart failure	1	0	0
Hypertension	1	1	1.3
Age ≥ 75	2	2	2.2
Diabetes	1	3	3.2
Stroke/TIA/thromboembolism	2	4	4.0
Vascular disease	1	5	6.7
Age 65–74	1	6	9.8
Sex category (female)	1	7	9.6
Maximum score	9	8	6.7
		9	15.2

AF = atrial fibrillation; TIA = transient ischemic attack.

Information from: January CT, Wann LS, Alpert JS, et al. 2014 AHA/ACC/HRS guideline for the management of patients with atrial fibrillation: a report of the American College of Cardiology/American Heart Association Task Force on Practice Guidelines and the Heart Rhythm Society. Circulation 2014;23:2071-104.

b. Anticoagulation decisions should balance the risk of stroke versus the risks of bleeding in the context of duration of bridging and/or lack of anticoagulation. Scoring systems have been described to help assess bleeding risk in anticoagulation decisions for patients with AF on warfarin anticoagulation (Table 10). Not unlike stroke risk scoring systems, these scoring systems were not founded in the context of critically ill patients.
c. Direct-acting oral anticoagulants (DOACs) are recommended over warfarin for patients with AF (exclusions: moderate to severe mitral stenosis, mechanical heart valve) (Chest 2018;154:1121-201; J Am Coll Cardiol 2019;12:104-32).
d. For patients with AF who have a CHA_2DS_2-VASc score of 2 or greater (men) and 3 or greater (women) and who have end-stage chronic kidney disease (creatinine clearance [CrCl] less than 15 mL/minute/1.73 m^2) or are receiving dialysis, it may be reasonable to prescribe warfarin (INR 2.0–3.0) or apixaban (J Am Coll Cardiol 2019;12:104-32).

Table 10. HAS-BLED or $HEMOR_2RHAGES$ Bleeding Risk Scores in AF

Risk Factor Assessment	Score	Total Patient Score	Bleeds/100 Patient-Yr of Warfarin
HAS-BLED			
Hypertension	1	0	1.13
Abnormal renal or liver function (1 point each)	1 or 2	1	1.02
Stroke	1	2	1.88
Bleeding	1	3	3.74
Labile INRs	1	4	8.70
Elderly (> 65 yr)	1	5	12.5
Drugs or alcohol (1 point each)	1 or 2	6	0
		Any score	1.56
$HEMOR_2RHAGES$			
Hepatic or renal disease	1	0	1.9
Ethanol abuse	1	1	2.5
Malignancy	1	2	5.3
Older age	1	3	8.4
Reduced platelet count or function	1	4	10.4
Rebleeding risk	2	≥ 5	12.3
Hypertension (uncontrolled)	1	Any score	4.9
Anemia	1		
Genetic factors	1		
Excessive fall risk	1		
Stroke	1		

AF = atrial fibrillation.

e. In the critically ill patient, parenteral anticoagulation with unfractionated heparin or low-molecular-weight heparin may be more favorable if the benefit of anticoagulation exceeds the risk of bleeding. Unfractionated heparin infusions would be favored more than low-molecular-weight heparin in renal failure (CrCl less than 30 mL/minute/1.73 m^2).

f. For anticoagulation of most critically ill patients, oral anticoagulation may be less favorable or even detrimental compared with parenteral anticoagulation.
   i. Elimination of DOACs (apixaban, dabigatran, edoxaban, and rivaroxaban) is significantly affected in renal compromise. An additional consideration in the critical care setting is the feasibility of reversal in the event of an acute bleed or need for an invasive procedure. Novel antidotes continue to be studied and approved (idarucizumab, andexanet); however, clinical experience with reversibility in these settings is evolving. Finally, the aforementioned DOACs are not devoid of relevant drug-drug interactions, particularly when given in combination with strong CYP and/or P-glycoprotein inhibitors/inducers.
   ii. Warfarin is challenging due to issues with malnutrition, drug interactions, and unpredictable dose response in the critically ill patient. Complications with this agent are more predictably managed than are complications with other oral anticoagulants if a patient requires an invasive procedure or develops an acute bleed, but warfarin use is still not without risk.

g. For patients with AF/flutter for 48 hours or more (or if undetermined) without therapeutic anticoagulation, absence of thrombus on the left side of the heart should be confirmed by transesophageal ECHO before pharmacologic or direct current cardioversion.

h. If AF/flutter for more than 48 hours or an unknown duration that requires direct current or pharmacologic cardioversion for hemodynamic instability, anticoagulation should be initiated as soon as possible and continued for at least 4 weeks after cardioversion unless contraindicated
i. For patients with AF/flutter taking anticoagulants who undergo cardiac stenting, the general recommended approach is to manage with dual therapy (anticoagulant + P2Y12 inhibitor). In some high ischemic risk patients, triple therapy with an anticoagulant + P2Y12 inhibitor + aspirin can be considered for up to 1 month after stenting. Dual therapy reduces the risk of bleeding compared with triple therapy without sacrificing ischemic efficacy. Further details surrounding recommendations are beyond the scope of this chapter but can be found in the 2023 ACC/AHA/ACCP/HRS Guideine for the Diagnosis and Management of Atrial Fibrillation Guidelines

Patient Case *(Continued)*

The appropriate intervention has been made from question 5, and the patient has not required any need for pacing. For the next 48 hours, J.M. continues on inotrope therapy and is being diuresed (his net fluid balance has been 1250 and 900 mL negative each day for the past 2 days). The transthoracic ECHO results showed J.M.'s ejection fraction to be 15%–20%, with a dilated, hypokinetic LV, severe mitral regurgitation, and dilated atria. No evidence of intracardiac thrombus was seen, but this could not be ruled out.

6. J.M. has had increasing premature atrial complexes on telemetry, and his heart rate has consistently been 83–96 beats/minute. He is receiving dobutamine at 5 mcg/kg/minute and a furosemide infusion at 10 mg/hour. The team is notified that J.M. has gone into AF with rapid ventricular response and a heart rate of 132 beats/minute (he has been in AF for about 30 minutes). His blood pressure is now 83/52 mm Hg. Which treatment plan would be most preferred for this patient?
   A. Synchronized cardioversion
   B. Metoprolol 5 mg intravenous push
   C. Adenosine 6 mg rapid intravenous push
   D. Amiodarone 300-mg intravenous push, followed by a continuous infusion at 1 mg/minute

7. J.M.'s BP and HR improved after the previous intervention; however, he remains in AF. The team is now concerned about evaluating the patient for anticoagulation. You are asked to provide input about the appropriateness of anticoagulation, given the patient's clinical course and past medical history. His calculated CHA_2DS_2-VASc score is 5 and HAS-BLED score is 5. The physician is considering anticoagulation with a heparin infusion while the patient is in the ICU and asks for a recommendation. Which is the most appropriate response?
   A. According to these scores, the patient's risk of bleeding is the same as his stroke risk; therefore, it would be reasonable to either initiate or withhold anticoagulation.
   B. According to these scores, the patient's risk of stroke exceeds his risk of bleeding; therefore, it would be reasonable to initiate heparin anticoagulation.
   C. The HAS-BLED score is less relevant because it is derived from chronic warfarin use. However, anticoagulation is still likely warranted eventually, given the patient's CHA_2DS_2-VASc score.
   D. The HAS-BLED and CHA_2DS_2-VASc scores are irrelevant to this patient's current care, and anticoagulation is not warranted.

VI. HEART FAILURE

A. Clinical Syndrome – Manifested because of congenital or acquired structural or functional myocardial dysfunction that impairs filling and/or emptying of the heart
   1. Mortality rate of 50% within 5 years of diagnosis
   2. Predominantly descriptive of LV and its pump performance and the ejection fraction
   3. Prognosis and treatment of HF largely depend on its etiology and staging.

Table 11. Diastolic Dysfunction Versus Systolic Dysfunction in HF

Type	Heart Failure with *Preserved* EF (HF*p*EF)	Heart Failure with *Reduced* EF (HF*r*EF)
Dysfunction	• Diastolic – blood filling the heart	• Systolic – blood emptying from the heart • May also coexist with impaired diastolic filling
Characteristics	• LVEF ≥ 50% • Impaired relaxation and filling of the ventricle before contraction • Commonly described as a "stiffened" ventricle • Contractility remains "normal"	• LVEF ≤ 40% • Impaired contraction and emptying of the ventricular cavity • Commonly described as a "dilated" ventricle
Clinical-management pearls	• Preload (volume optimization) is essential • As diastolic dysfunction worsens – ventricular preload and diastolic filling time must be maintained • Conditions acutely decreasing preload may lead to decompensation (e.g., tachyarrhythmias) • Keep heart rate "slow" and ventricles "full"	• Preload optimization essential – at high risk of volume overload • As systolic dysfunction worsens, cardiac output becomes increasingly afterload sensitive • Conditions acutely decreasing or increasing preload or increasing afterload may lead to decompensation • Volume overload and increasing myocardial dilation may contribute to decreased valvular coaptation and regurgitant flow
Common medications to avoid	• NSAIDs, with the exception of aspirin • Sympathomimetics	• NSAIDs, with the exception of aspirin • Sympathomimetics • Most antiarrhythmics except for amiodarone and dofetilide • Non-dihydropyridine calcium channel blockers

EF = ejection fraction; LVEF = left ventricular ejection fraction.

B. HF Etiologies
   1. Ischemic cardiomyopathy
      a. Accounts for about two-thirds of HF cases
      b. Caused by myocardial damage/death owing to CAD
   2. Non-ischemic cardiomyopathy
      a. Accounts for about one-third of HF cases
      b. May be attributed to other causes (Box 4)

Box 4. Non-ischemic Cardiomyopathies

Non-ischemic Cardiomyopathies	
• Amyloidosis • Sarcoidosis • Drug induced ◦ Alcohol ◦ Anabolic steroids ◦ Chemotherapy § Anthracyclines § Cyclophosphamide § Fluorouracil § Bevacizumab § Trastuzumab § Tyrosine kinase inhibitors • Cocaine • Methamphetamine • Tricyclic antidepressants • Anorexigens	• Familial • Hypertension • Hyper/hypothyroidism • Hypertrophic obstructive (HOCM) • Idiopathic • Myocarditis ◦ Autoimmune ◦ Eosinophilic ◦ Giant cell ◦ Viral ◦ Other infectious cause • Non-compaction • Peripartum • Obesity • Stress induced (Takotsubo)

C. RV Dysfunction and/or Failure (Exp Clin Cardiol 2013;18:27-30; J Am Coll Cardiol 2010;56:1435-46; Circ Heart Fail 2010;3:340-6; Anesth Analg 2009;108:422-33; Anesth Analg 2009;108:407-21)
   1. Less well characterized than LV HF
   2. No guidelines to direct management of acute RV failure
   3. RV physiology
      a. The RV normally has only about one-sixth the myocardial mass of the LV. Primary means of compensation is heart rate increase.
      b. Normal RV ejection fraction is 40%–45%.
      c. Conduit to a low pressure system in the pulmonary vasculature (normally) and capable of accommodating for increased stroke volume but not able to compensate for acute pressure fluctuations.
      d. Physiologic deficits:
         i. Preload dependent
         ii. Interdependent on LV and septal contribution to contraction
         iii. Highly sensitive to acute increases in afterload. PVR elevations may be:
            (a) Drug induced (e.g., α_1 agents, protamine)
            (b) Caused by ventilator settings (high positive end-expiratory pressure [PEEP], high tidal volumes)
            (c) Caused by hypoxia
            (d) Caused by hypercarbia
            (e) Caused by pulmonary embolism
         iv. Poor elastic response to acute preload increases compared with LV
         v. Atypical geometric form contributes to:
            (a) Difficulty in objective assessment of RV function by ECHO
            (b) Increases susceptibility to further inefficiency/dysfunction when severely dilated

4. Etiologies of RV failure

Box 5. Etiologies of RV Failure (Exp Clin Cardiol 2013;18:27-30; J Am Coll Cardiol 2010;56:1435-46)

ARDS
Arrhythmias
Cardiac tamponade
Congenital heart disease
Heart transplantation (particularly if prolonged ischemic time)
Hypovolemia
Hypoxia
LV dysfunction
Mitral valve disease
Post-cardiac surgery
Pulmonary embolism
Pulmonary hypertension
Pulmonary regurgitation
Right coronary artery infarction/ischemia
RV overload
Sepsis
Tricuspid regurgitation/stenosis

D. Diagnostic Tests for New or Worsening HF – In addition to routine chemistry and CBC tests, additional testing may include:
   1. Liver function tests (may be indicative of congestive hepatopathy, if elevated)
   2. 12-lead ECG
   3. Troponin
   4. Left heart catheterization if suspected new ischemic contribution
   5. BNP or N-terminal pro-brain natriuretic peptide (elevations may help in the diagnosis of acutely decompensated HF in scenarios of uncertainty)
   6. Transthoracic or transesophageal ECHO
   7. Invasive hemodynamic monitoring (to guide volume optimization and dosing response to inotropes or vasopressors)
   8. For less common cardiomyopathies, noninvasive imaging (i.e., cardiac magnetic resonance imaging [MRI]) and/or myocardial biopsy may be required.

E. General Management Considerations
   1. Assess volume status.
      a. In selected cases of hypotension where HF is not known to be the exclusive culprit, small volume fluid challenges (250–500 mL intravenous fluid bolus) or passive leg raising (PLR) maneuvers may help show whether a patient is volume responsive.
      b. PLR is a reliable and alternative way of evaluating volume status, without the need to administer a fluid challenge (Curr Opin Crit Care 2017;23:237-43).
      c. Physical examination will generally guide fluid status decision.
   2. Once other causes of symptomatic congestion and/or low perfusion at rest have been ruled out and a diagnosis of decompensated HF is deemed likely, the following should be considered: If volume overloaded, provide intravenous diuretics at least equal to the patient's home oral dose (or alternatively up to 2–2.5 times the patient's home dose may be used). For patients who are diuretic naive on admission, a starting dose of 20–80 mg of furosemide (or equivalent) is reasonable.

a. Monitor for clinical improvement in symptoms and adequate urine output.
b. Proactive evaluation of urine output adequacy should be used to facilitate optimization/escalation of diuretic regimen.
c. Increase in loop diuretic dose and/or addition of sequential nephron blockade (i.e., metolazone) may be considered as needed to achieve the desired urine output and fluid balance. Recent evidence also suggests that SGLT2 inhibitors and acetazolamide may also help improve response to diuretic therapy (Circulation 2022;146:289-98; N Engl J Med 2022;387:1185-95). Administration of the loop diuretic as a continuous infusion can also be considered.
d. For patients unresponsive to diuretic escalation as above or those with end-stage renal disease, hemodialysis and isolated ultrafiltration can be used for fluid removal.

3. Evaluate for use of vasodilator therapy.
   a. If normotensive or hypertensive (e.g., SBP greater than 100 mm Hg) in the presence of acute pulmonary edema (despite diuretic therapy):
      i. Continuous infusion intravenous vasodilators (e.g., nitroglycerin or nitroprusside) should be considered.
      ii. If the presence of high afterload is confirmed (systemic vascular resistance), then nitroprusside may be warranted more than other vasodilators. However, careful attention to patient selection is important because nitroprusside elimination is dependent on both hepatic metabolism and renal clearance. Note that nitroprusside is converted to nitric oxide and cyanide, with subsequent conversion to thiocyanate. Signs of potential toxicity include metabolic acidosis and mental status changes. In addition, patients receiving nitroprusside should have an arterial line for close blood pressure monitoring and typically a Swan-Ganz catheter in place.
      iii. Pulmonary vasodilators may be required in patients with evidence of pulmonary arterial hypertension: inhaled nitric oxide, epoprostenol, or alternatively phosphodiesterase type 5 inhibitors (sildenafil). Inhaled nitric oxide and epoprostenol may also be considered as short-term adjuncts in patients with right ventricular dysfunction and pulmonary hypertension in the perioperative setting. The delivery system and high cost associated with inhaled nitric oxide is a limitation relative to other options.
   b. If hypotensive (e.g., SBP of 100 mm Hg or less): Can consider vasodilators, but with caution
4. Monitor for clinical improvement of symptoms and adequate urine output.
   a. Proactive evaluation of urine output adequacy should be used to facilitate optimization/escalation of diuretic regimen. An optimized diuretic dose should produce 100–150 mL/hour of urine output at 6 hours or 50–70 mEq/L of urine sodium at 2 hours.
5. Evaluate for use of inotrope therapy
   a. If inadequate response on escalation of diuretics (and vasodilators, if appropriate), inotrope therapy should be considered, accounting for any concurrent physiologic considerations (e.g., right HF, pulmonary hypertension, ischemia, or valvular disease)
   b. Dobutamine and milrinone are most often used for suspected or confirmed low cardiac output states. Milrinone may be preferred more than dobutamine in the presence of recent administration of β-blockade or in the setting of concomitant pulmonary hypertension (because of its post-β receptor effects and potent vasodilatory properties, respectively). In terms of pharmacokinetic differences, dobutamine has a faster onset and shorter half-life (2 min) compared with milrinone (half-life of 3 hrs with renal-dependent elimination). However, no differences in clinical efficacy or safety have been observed between agents. Both agents should be closely monitored for the development of proarrhythmias and worsening ischemia.

6. Evaluate for the appropriate initiation of GDMT
    a. Current guidelines recommend initiation of a 4-drug regimen for most patients with HFrEF as soon as possible, including in the inpatient setting. This 4-drug regimen is a combination of a RAAS inhibitor (angiotensin receptor/neprilysin inhibitor preferred), beta blocker, MRA, and sodium glucose co-transporter 2 inhibitor. For further guidance on GDMT indications, please consult current HF guidelines (Circulation 2022;145:e895-1032).

VII. VALVULAR HEART DISEASE

A. Valvular heart disease is an important comorbid condition that must be considered in hemodynamic management of the critically ill patient. Valvular heart disease can also independently lead to the presenting critical illness.

B. Among the four cardiac valves, several etiologies[a] contribute to the manifested pathology; however, the depth and breadth of these etiologies are beyond the scope of this chapter. Nevertheless, conditions that may require repair/replacement include those listed in Table 12.

Table 12. Conditions That May Require Valvular Repair/Replacement

Stenosis	Narrowing at the opening of the valve(s) Can lead to concurrent regurgitation
Regurgitation	"Leaky" valve(s) resulting in less blood pumping forward through the heart
Prolapse	"Floppy" valve(s) with part of valve not working
Endocarditis	Infection of one or more valves
Malformation	Often occurs at birth when the valve (or valves) is defective

[a]Valvular disease secondary to rheumatic heart disease is a rare but possible contributor.

Table 13. Valvular Disease Characteristics and Management Considerations[a,b] (Circulation 2021;143:e72–227)

Valvular Disease Type	Management Considerations
Aortic stenosis (AS)	• One of the most common and serious valvular diseases seen in the ICU • As stenosis and disease progresses, severe/critical AS limits the heart to a fixed stroke volume; this inability to increase stroke volume occurs despite intrinsic or extrinsic attempts to compensate (i.e., increased chronotropy or inotropy) and often only increases myocardial oxygen demand without improving delivery • Must be extremely cautious in approaching and reacting to invasive hemodynamic variables (particularly cardiac output and cardiac index) – adding agents with inotropic/chronotropic effects may expedite demand ischemia and an acute MI because of the physiologic inability to increase cardiac output with a fixed partial LV outflow tract obstruction • For reasons similar to those previously listed, treating hypertension with afterload-reducing agents must be done judiciously because as afterload (SVR) decreases, cardiac output cannot increase • Can coexist with concurrent aortic insufficiency/regurgitation because the stenotic or calcified valve leaflets may no longer move and come together (coapt) well • These patients may be at risk of developing mitral regurgitation and subsequent increases in left atrial pressures (increasing risk of AF) because of increased LV filling pressures • Must be cautious in decreasing HR in sinus tachycardia because this can be a primary means of compensation • Patients need adequate preload; however, they can become symptomatic even with slight volume overload. As an example, atrial fibrillation can be very detrimental simply because it decreases preload to the ventricle • Patients with severe aortic stenosis and systolic HF (described as low output – low gradient) have a poorer prognosis
Aortic regurgitation (AR)	• Also known as aortic insufficiency (AI) • Must be cautious in decreasing HR in sinus tachycardia because this can be a primary means of compensation to maintain adequate cardiac output • Patients need adequate preload, but the predominant target is to maintain decreased afterload (SVR) to facilitate forward blood flow and cardiac output
Mitral stenosis (MS)	• Contributes to decreased LV filling and increased left atrial pressures; can increase risk of AF and secondary pulmonary hypertension • Increasing diastolic filling time and avoiding tachycardias can facilitate stabilization until valve is corrected • Use of selective pulmonary vasodilators (i.e., inhaled nitric oxide or inhaled epoprostenol) may be detrimental because these agents can facilitate pulmonary congestion, given the preexisting pulmonary venous hypertension and elevated left atrial pressures • Can coexist with concurrent mitral regurgitation because the stenotic or calcified valve leaflets may no longer move and coapt well
Mitral regurgitation (MR)	• Primary clinical target is to decrease LV afterload (SVR) to minimize augmentation of MR and to facilitate forward blood flow. If SVR is too high, blood will travel in the path of least resistance until an adequate LV pressure is generated to open the aortic valve (must exceed the systemic diastolic blood pressure) • Use of selective pulmonary vasodilators (i.e., inhaled nitric oxide or inhaled epoprostenol) may be detrimental because these agents can facilitate pulmonary congestion, given the preexisting pulmonary venous hypertension and elevated left atrial pressures

Table 13. Valvular Disease Characteristics and Management Considerations[a,b] (Circulation 2021;143:e72–227) *(continued)*

Valvular Disease Type	Management Considerations
Tricuspid regurgitation (TR)	• Likely to influence pulmonary artery catheter assessments of cardiac output by way of thermodilution technique • Can be influenced by infectious causes and presence of indwelling transvenous catheters or leads; but moderate or severe TR is more commonly a marker of RV overload and dysfunction

[a]Infective endocarditis can cause progressive valve disease, leading to regurgitant flow and impaired valve leaflet coaptation; however, it can also lead to near obstruction in some cases.

[b]Degree of valvular disease is graded as mild, moderate, or severe, as defined by objective ECHO or catheter-based assessments.

C. Procedural or Surgical Correction

Table 14. Procedural or Surgical Correction (Circulation 2017; Available at http://circ.ahajournals.org/content/early/2017/03/14/CIR.0000000000000503)

Balloon valvuloplasty	• Performed by percutaneous intervention as a temporizing intervention
Valve repair	• May entail direct surgical repair of a damaged valve leaflet or implantation of a ring at the valve annulus to facilitate improved coaptation of a regurgitant valve
Tissue (bioprosthetic) valve	• Made of animal or human tissue • Usually does not require long-term anticoagulation (see Table 15) • Does not last as long as a mechanical valve (may last 10–15 yr)
Mechanical valve	• Made of synthetic materials (newer valves use ceramic or carbon) • They are durable and generally unlikely to need replacement • Require lifelong anticoagulation • Warfarin is currently the only anticoagulant approved for use by the FDA in patients with mechanical heart valves (see Table 15)
Transcatheter aortic valve replacement (TAVR)	• Made of animal tissue (bioprosthetic) and attached to a wire frame stent and placed using catheter inside the old aortic valve • This may be considered in patients who are at higher perioperative risk for surgical aortic valve replacement or when surgery is not an option • Anticoagulation and/or antiplatelet agents are required for at least a short time after TAVR
Transcatheter edge-to-edge repair (TEER)	• Minimally invasive technique for treatment of symptomatic chronic moderate-severe or severe primary (degenerative) mitral regurgitation in patients at prohibitive surgical risk • A leaflet repair device (MitraClip) is currently the only FDA approved device for this indication • This device uses a cobalt chromium clip to suture the regurgitant orifices of the mitral valve leaflets together, thereby increasing coaptation • Anticoagulation and/or antiplatelet agents are required for at least a short time after TEER

D. Anticoagulation
   1. Three guidelines exist regarding valve anticoagulation, with varying agreement in the recommendations.
   2. Prosthetic mitral valves have increased risk of thrombosis (blood flow across the valve is passive and occurs during diastole) versus aortic valves, where blood flow across the valve is active occurring during systole.

Table 15. Summary of Anticoagulation Recommendations for Patients with Prosthetic Valves (Circulation 2017; Available at http://circ.ahajournals.org/content/early/2017/03/14/CIR.0000000000000503)

Procedure	Warfarin, Target INR Range	Class of Recommendation/ Level of Evidence[a]	Antiplatelet Therapy	Class of Recommendation/ Level of Evidence[a]
Mechanical AVR	2–3 (bileaflet or current-generation single-tilting disc and no risk factors for thromboembolism[b])	IB	Aspirin 81 mg daily may be considered if indicated for antiplatelet therapy and bleeding risk is low	IIb/B
	2.5–3.5 (older-generation [i.e., ball-in-cage valve, or with additional risk factors for thromboembolism[b]])	IB		
	For a mechanical On-X AVR and no thromboembolic risk factors, a goal INR of 1.5–2.0 may be reasonable starting ≥3 mo after surgery	IIb/B		
Bioprosthetic AVR	2–3 for ≥ 3 mo (and up to 6 mo if low bleed risk)	IIa/B	Aspirin 81 mg daily may be considered if indicated for antiplatelet and bleeding risk low	IIb/B
Transcatheter AVR	2–3 for ≥ 3 mo if low bleed risk	IIb/B	Alternative to warfarin: Clopidogrel 75 mg daily for 6 mo and aspirin 81 mg daily	IIb/B
Mechanical MVR	2.5–3.5	I/B	Aspirin 81 mg daily may be considered if indicated for antiplatelet therapy and bleeding risk is low	IIb/B
Bioprosthetic MVR	2–3 for ≥ 3 mo (and up to 6 mo if low bleed risk)	IIa/B	Aspirin 81 mg daily	IIa/B

[a]Atrial fibrillation, previous thromboembolism, left ventricular dysfunction, or hypercoagulable conditions.

[b]Class of Recommendations: I = evidence and/or general agreement that a given treatment or procedure is beneficial, useful, effective; II = conflicting evidence and/or a divergence of opinion about the usefulness/efficacy of the given treatment or procedure; IIa = weight of evidence/opinion is in favor of usefulness/efficacy; IIb = usefulness/efficacy is less well established by evidence/opinion; III = evidence of general agreement that the given treatment or procedure is not useful/effective and, in some cases, may be harmful; Levels of Evidence: A = data derived from several randomized clinical trials or meta-analyses; B = data derived from a single randomized trial or large nonrandomized studies; C = consensus of opinion of the experts and/or small studies, retrospective studies, registries.

[c]Target INR 2-3 for the first 3 mo.

E. Valve Thrombosis
   1. Highest risk within the first year after replacement
   2. Most predominant risk is in mechanical prosthetic valves, but can also occur with bioprosthetic valves
   3. Treatment involves systemic fibrinolytic treatment or surgical reoperation with varying success rates – Both carry significant risks of morbidity and mortality, including:
      a. Thrombolysis (predominantly recommended for right-sided valve thrombosis)
         i. Cardiac tamponade
         ii. Stroke/transient ischemic attack
         iii. Systemic embolization
         iv. Major bleeding
         v. Death
      b. Surgery (predominantly recommended for left-sided valve thrombosis)
         i. Cardiac tamponade
         ii. Stroke/transient ischemic attack
         iii. Renal failure
         iv. Heart block
         v. Systemic embolization
         vi. Prolonged ventilation
         vii. Major bleeding
         viii. Death

F. Other Cardiac Functional Defects
   1. Left ventricular outflow tract (LVOT) obstruction (Circulation 2020;142:e558-e631; Circulation 2008;117:429-39; J Am Coll Cardiol 2000;36:1344-54)
      a. Hypertrophic obstructive cardiomyopathy (HOCM)
         i. Genetic disease leading to hypertrophy of the LV (particularly the ventricular septum) with or without the presence of LV outflow tract obstruction
         ii. LVOT obstruction is of greatest concern when HOCM exists with systolic anterior motion (SAM) (see text that follows); however, in HOCM, LVOT obstruction can exist in the absence of SAM because of septal hypertrophy; nonetheless, acute clinical management is predominantly the same as outlined in 1.a.i–iii and in 1.b.i.
         iii. Treatment of HOCM without SAM relies heavily on the management of contributors to myocardial hypertrophy and diastolic HF.
            (a) Non-vasodilating β-blockers are first-line agents in the treatment of patients with HOCM and are commonly titrated to heart rate goals of 60–65 beats/minute.
            (b) Dihydropyridine calcium channel blockers (particularly verapamil) are recommended in patients with contraindications to β-blockade or in those without advanced HF or bradycardia.
            (c) For recommendations specific to HOCM (patients also with SAM), see "b" in the text that follows.
            (d) Patients with HOCM may be at increased risk of sudden cardiac death. Evaluation should consider the patient's candidacy for implantable cardioverter-defibrillator placement by ambulatory ECG (Holter) monitoring at least biannually.
         iv. Treatment may be indicated by surgical septal myectomy or catheter-directed alcohol ablation.
      b. SAM of the mitral valve
         i. SAM and LVOT obstructions result in a systolic outflow tract obstruction and are commonly associated with HOCM but can also occur in other clinical scenarios, particularly after cardiac surgery.

ii. Can lead to severe cardiogenic shock
iii. SAM is more of a dynamic obstruction in which the degree of obstruction and flow gradient is dependent on heart rate, cardiac contractility, and ventricular preload volume.
(a) In an underfilled LV, there is physically less distance between the mitral valve and septum, thus generating an increased risk of obstruction because the LVOT is generally narrower at the onset of systole, particularly if the mitral valve leaflet is affected.
(b) Increasing cardiac contractility and heart rate increases LVOT obstruction and gradient by inducing a stronger contraction, increasing the contact between the septum and mitral leaflets, and increasing the rate of systolic attempts.
iv. For patients with SAM who have a potential for obstructive physiology, management involves maintaining normal or increased LV preload and low heart rates.
(a) Acute hypotension is best managed with phenylephrine or vasopressin (pure vasoconstrictors) to selectively increase SVR without increasing contractility or heart rate. Concomitant β-blocker use may also be considered to improve cardiac filling.
(b) Inotropes and vasopressors that mediate increases in heart rate or contractility should be avoided, if possible, because they may be harmful and worsen the LVOT.
(c) Afterload-reducing agents (e.g., ACE inhibitors, ARBs, non-dihydropyridines) should be used with caution (if at all).

2. Septal defects (atrial or ventricular)
a. Septal defects can be acquired (i.e., postinfarction ventricular septal defect) or can be congenital.
b. Diagnosed predominantly by ECHO using a bubble study. If the patient presents in a seemingly low cardiac output state, a left-to-right intracardiac shunt should be suspected if mixed venous oxygen saturation (Svo_2) saturations are greater than $Scvo_2$ saturations.
c. Important principles
i. Goals include minimizing the degree of intracardiac shunt while maintaining adequate cardiac output. It is generally favored to accept a right-to-left intracardiac shunt while recognizing that partial pressure of arterial oxygen ($Pao_{2)}$ saturations will be somewhat decreased and reflective of venous and arterial blood mixing in the LV before ejection.
ii. Decreasing right-sided cardiac filling pressures can augment left-to-right intracardiac shunting of blood. Administration of venodilators (i.e., nitrates) or aggressive diuresis could augment left-to-right intracardiac shunts and lead to clinical deterioration.
iii. Intravenous medications should preferably be filtered to minimize the risk of air/particulate embolus traveling through to the left side of the heart, being ejected, and causing a potential stroke.
d. Treatment may include surgical correction or percutaneous catheter placement of a closure device.

VIII. ADVANCED THERAPIES FOR HEART FAILURE AND CARDIOGENIC SHOCK

A. The goals behind advanced therapies can be thought of as dynamic, depending on patient progression and resolution or presentation of comorbid conditions.

B. Dynamic Progression of HF Advanced Therapies

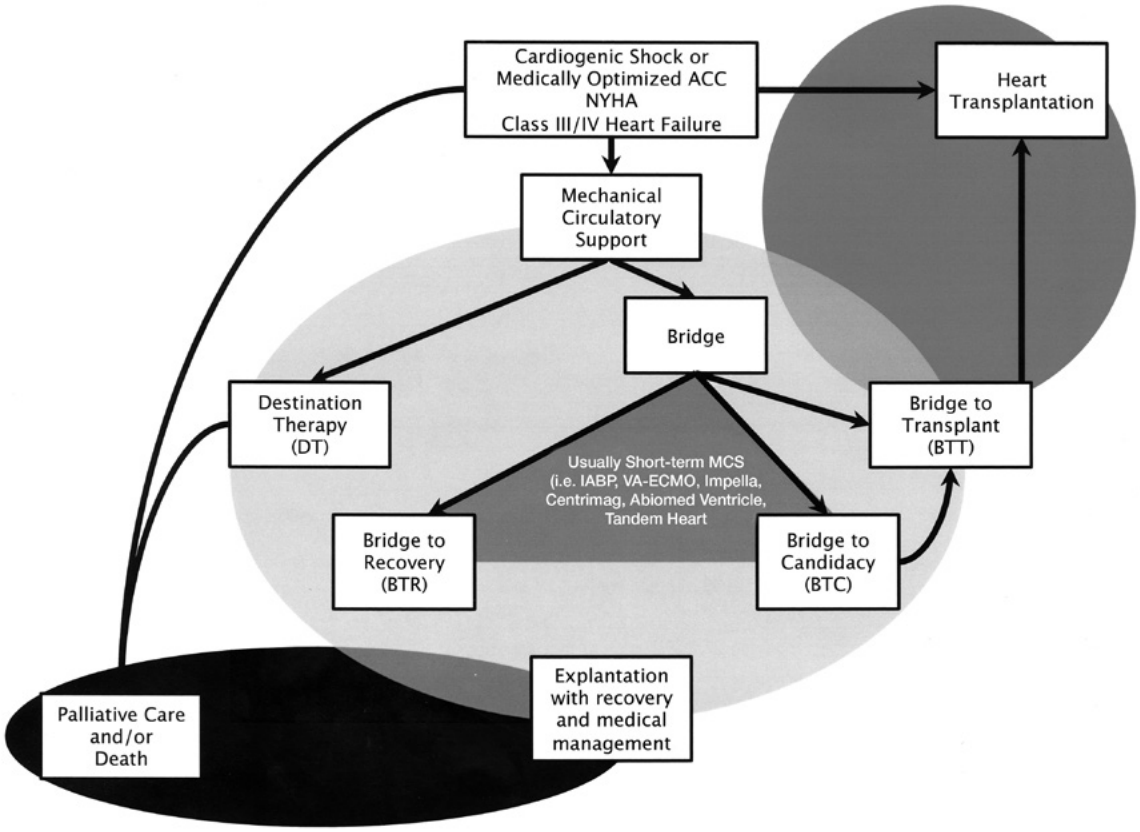


Figure 10. Dynamic progression of heart failure advanced therapies.

C. Mechanical Circulatory Support (MCS) (J Heart Lung Transplant 2013;32:157-87)
   1. Intent, duration, and type of MCS support depend on several factors, including patient acuity, comorbidities, and prognosis.
   2. Common terms
      a. Destination therapy – Formal designation for patients who meet the criteria for long-term mechanical support but who are not a transplant candidate because of relative or absolute contraindications
      b. Bridge to transplantation – Formal designation for patients eligible to be listed as candidates for heart transplantation
      c. Bridge to candidacy and bridge to recovery – Used in concept and are not formally recognized abbreviations but describe the approach to temporary MCS when short-term left ventricular assist devices (LVADs) are used to support a patient until a long-term prognosis can be determined, which may include explantation with recovery, implantation of long-term durable LVAD support, heart transplantation, or palliative care

Table 16. Mechanical Interventions for Shock

Short to Intermediate Term	
Intra-aortic balloon pump (IABP) counterpulsation	• Placed by femoral arterial catheter or percutaneously through the axillary artery and advanced up to the aorta • Inflation enables diastolic augmentation of systemic blood pressure to improve vital organ and coronary perfusion pressures • Deflation facilitates selective afterload during systole through a vacuum effect of the balloon deflation to ease cardiac output (does not technically increase cardiac output) • Can be set to trigger from ECG, pacer, or arterial line pressure, or can be manually set • Tachyarrhythmias and aortic regurgitation/insufficiency are not well supported with this means of MCS • Level of support coincides with timing of inflation/deflation per related heartbeat; for example: 1:1 = one inflation/deflation per every heartbeat (maximal support) 1:4 = one inflation/deflation for every fourth heartbeat (less support) • When setting duration in deflated state increases (providing less support), the thrombosis risk associated with the IABP increases, commonly requiring anticoagulation • Complications may include limb and visceral ischemia (distal pulses must be monitored three times a day), thrombocytopenia, hemolysis, balloon rupture.
Percutaneous VAD	• One common example is Impella: intraluminal axial support that provides varying degrees of LV output support (CP and 5.5) and RV output support (RP) • Another example is TandemHeart: left atrium-to-femoral artery bypass using transseptal cannulation • May be considered in cardiogenic shock or as temporary support during high-risk PCI • Anticoagulation regimen is a common topic of debate and medication safety discussion. In general, UFH is accepted as the standard anticoagulant for percutaneous VADs, and bivalirudin and argatroban can be used in patients with a history of or concern for HIT. Sodium bicarbonate-based purge solution is an emerging alternative for patients with Impellas who cannot receive heparin (Pharmacotherapy 2021;41:932-42; Ann Pharmacother. Published online September 13, 2022). This purge solution should generally be used in combination with systemic anticoagulation with a previously mentioned agent. • Complications may include hemolysis/bleeding, arrhythmias, and migration and/or malposition of the catheter/cannula
Extracorporeal or paracorporeal VAD	Examples include CentriMag, BVS 5000, and AB5000 ventricle

Table 16. Mechanical Interventions for Shock *(continued)*

Short to Intermediate Term	
Extracorporeal life support (ECLS) or extracorporeal membrane oxygenation (ECMO)	• Similar to cardiopulmonary bypass in which large-bore cannulas drain venous blood that is pumped through an oxygenator, where it is oxygenated and/or cleared of carbon dioxide and then actively pumped back into the body • Modality of support depends on the means of vascular cannulation ***Venoarterial:*** Removal of venous blood from the vena cava with circulation through the ECMO circuit and delivery in retrograde fashion up the aorta; potentially indicated for primary cardiogenic shock, cardiopulmonary failure, and post-cardiopulmonary circulatory shock ***Venovenous:*** Removal of venous blood from the vena cava with circulation through the ECMO circuit and delivery back to the right atrium; potentially indicated for hypoxic respiratory failure owing to any cause, hypercarbic respiratory failure with bronchospastic disease or other cause of carbon dioxide (CO_2) retention, or severe air leak syndromes
Long term	
Implantable LVADs	Examples include the Heartmate II, HeartWare, Heartmate III
Total artificial heart	Example includes Syncardia TAH (biventricular support)

LVAD = left ventricular assist device; VAD = ventricular assist device.

d. Anticoagulation considerations
   i. Anticoagulation strategies are specific to the proprietary device, and many institutions have standardized protocols.
   ii. Safety and efficacy of DOACs in patients with ventricular assist devices (VADs) have not been well established.

3. Complications of MCS (other than device failure)
   a. Bleeding
      i. Common sources
         (a) Nasal/upper airway
         (b) Gastrointestinal (GI)
         (c) Arteriovenous malformations in one of the previously stated locations
         (d) Hemolysis
      ii. Workup and/or acute treatment options
         (a) Laboratory workup
            (1) Prothrombin time/INR/aPTT
            (2) Increase frequency of hemoglobin/hematocrit evaluation.
            (3) Multimeric von Willebrand testing for acquired von Willebrand factor deficiency (some clinicians believe that all patients with prolonged continuous flow MCS develop acquired von Willebrand disease)
            (4) If no overt sign of bleeding – Consider hemolysis workup.
         (b) If suspected/confirmed bleeding, hold anticoagulation and consider reversal with caution. Consider any history of bleeding/clotting-related problems and indications for antithrombotic therapy in addition to MCS.
         (c) Obtain appropriate consults, and consider common interventions.
         (d) Ear, nose, and throat:
            (1) Evaluate for source control and/or cauterization.

(2) Mupirocin 2% (Bactroban) every 12 hours to each nostril to maintain moist nasal passages
(3) Oxymetazoline 0.05% (Afrin) to each nostril every 12 hours as needed for epistaxis

(e) Gastroenterology
(1) Proton pump inhibitor continuous infusion until location of GI source identified or for up to 72 hours. Intermittent intravenous dosing may also be considered.
(2) Enteroscopy with or without colonoscopy; may consider one or more of the following:
- Angiography and cautery
- Balloon-assisted enteroscopy
- Video-capsule enteroscopy
- Surgery

iii. Long-term management
(a) Consider decreasing anticoagulation/antiplatelet therapy intensity.
(b) Refractory bleeding despite previously stated interventions
(1) Consider role of oral antifibrinolytics (i.e., aminocaproic acid).
(2) Consider role of desmopressin or von Willebrand factor replacement if confirmed acquired von Willebrand disease and if LVAD speed cannot be further decreased.
(3) If GI arteriovenous malformations are present that are not amenable to intervention, octreotide 50 mcg subcutaneously every 8 hours may be considered.

b. Hemolysis and/or thrombosis
i. Hemolysis may be the presenting symptom of an underlying process, including infection, pump thrombosis, or other mechanical or physiologic dysfunction.
ii. Common presentation includes:
(a) Nonhemorrhagic anemia
(b) Urine color changes with appearance of hematuria; in severe cases, can be brown or black
(c) Hyperkalemia
(d) LVAD pump alarms
iii. Workup for contributing factors:
(a) Evaluate LVAD and cardiac function for contributing factors, including documentation and alarm history for suction events, power spikes, speed changes, volume status, RV function, arrhythmias
(1) Evaluate cannula(e) position and evaluate for obstruction/thrombus by ECHO or computed tomography (CT).
(2) Evaluate RV function by ECHO.
(3) Evaluate for intra-device thrombosis by ECHO ramp testing (Echocardiography 2014;31:E5-9; J Am Coll Cardiol 2013;62:2149-50; J Am Coll Cardiol 2012;60:1764-75).
(b) Ensure adherence to anticoagulation regimen according to patient's established goals.
(c) Evaluate laboratory values to establish the presence and degree of hemolysis and to identify any other potential contributors.
(1) Lactate dehydrogenase (LDH); normal values are 300–600 IU/L for most forms of MCS.
(2) Haptoglobin
(3) Plasma-free hemoglobin
(4) Blood cultures (an association has been identified with bacteremia and hemolysis in patients with LVADs – in some cases, presenting as thrombosis).

(5) Urine assessment – Send baseline urinalysis, urine culture, and appearance of changes daily to assess for improvement. Likely hemolysis, as evidenced by the presence of casts and color changes (darkened tea, red, or black are highly suggestive of hemolysis).

(d) Other laboratory values/considerations
(1) Hypercoagulable states (i.e., heparin-induced thrombocytopenia [HIT])
(2) CBC with differential
(3) Reticulocyte count

iv. Treatment
(a) Address any evidence of mechanical dysfunction by adjusting speed/flow rates, if possible.
(b) Consider medical optimization of RV function.
(c) If hypovolemic, give volume challenge. Consider alkalinization of urine and optimize fluid status (sodium bicarbonate 150 mEq/1000 mL of sterile water) at 0.5–1 mL/kg/hour; treat to a goal urine pH of greater than 7.5 to avoid additional hemolysis/hemoglobinuria-related acute kidney injury.
(d) Evaluate and optimize anticoagulation strategy.
(1) Ensure therapeutic anticoagulation with heparin or warfarin.
(2) Antithrombotic therapy that is more aggressive may be appropriate. Optimal acute antithrombotic strategies, although not yet defined, may include heparin infusion, glycoprotein (GP) IIb/IIIa infusions, parenteral direct thrombin inhibitors, or thrombolytics.
(3) Severe hemolysis can potentiate platelet activation – Can consider GP IIb/IIIa antagonist therapy (see Table 5), depending on bleeding risks and potential surgical plan.
(e) Reassess long-term antithrombotic strategy.
(1) Consider augmentation of antiplatelet therapy, or increase the INR therapeutic goal range.
(2) If thought to be related to a concurrent infection, can consider acutely increasing anticoagulation goals until infection control is gained

c. Infection
i. LVAD infections are often complex and have been characterized by the International Society for Heart & Lung Transplantation in the following manner (J Heart Lung Transplant 2011;30:375-84):
(a) VAD-specific infections
(1) Pump and/or cannula infections
(2) Pocket infections
(3) Percutaneous driveline infections
- Superficial infection
- Deep infection

(b) VAD-related infections
(1) Infective endocarditis
(2) Bloodstream infections that may be VAD related or non–VAD related
(3) Mediastinitis
- Sternal wound infection: Surgical site infection-organ space
- VAD pocket infection (continuous with mediastinum or already situated in the mediastinum, depending on the device used)
- Other causes of mediastinitis, perforation of the esophagus

(4) Non-VAD infections
- Lower respiratory tract infection
- Cholecystitis
- *Clostridium difficile* infection
- Urinary tract infection

ii. Antibiotic treatment
(a) Antibiotic coverage should account for site of suspected infection, previous pathogens and susceptibilities, proximity to driveline site or VAD pocket, and any other potential exposure or bacteremia secondary to the procedure. Prophylactic antibiotic coverage for VAD-related or non–VAD-related surgical procedures should account for site of procedure, previous infections, proximity to driveline site or VAD pocket, and any other potential exposure or bacteremia secondary to the procedure. In some circumstances, this requires broader prophylactic antibiotic coverage (Interact Cardiovasc Thorac Surg 2012;14:209-14).
(b) Treatment duration depends largely on the type of infection. However, if the infection is VAD related or VAD specific, prolonged antimicrobial therapy (more than 4 weeks) is commonly used.
(c) Because LVAD exchange is not without considerable risk, long-term oral antibiotic suppression therapy may be considered for some infections.

Patient Case *(Continued)*

8. J.M. is no longer in AF but remains in cardiogenic shock. The cardiac intensive care unit team has consulted the cardiothoracic surgeons for evaluation of his mitral valve disease and has considered advanced HF therapies. In the patient's decompensated state, he would likely need additional optimization if he were to undergo surgery. Which temporary means of mechanical circulatory support (MCS) might be most favorable to help stabilize this patient's cardiogenic shock in the setting of moderate to severe mitral regurgitation?
   A. Venoarterial extracorporeal membrane oxygenation (ECMO)
   B. Venovenous ECMO
   C. Intra-aortic balloon counterpulsation
   D. None; the patient likely requires urgent surgery.

D. Heart Transplantation (J Heart Lung Transplant 2010;29:914-56)
1. Transplantation remains the gold-standard treatment for end-stage HF.
2. Many variables limit the utility of heart transplantation, foremost of which is donor availability and donor-recipient tissue compatibility. These limitations (and potentially others) may make long-term MCS a more viable option until heart transplantation becomes an option.
3. Other limitations may include recipient characteristics of:
   a. Mental health
   b. Social support
   c. Adherence to medication and appointments
   d. Severe pulmonary hypertension
   e. Cancer
   f. Infection
   g. Tobacco or ethanol use
   h. Illicit drug use history
4. Additional considerations associated with thoracic transplantation in the ICU, such as immunosuppression, rejection, and other complications, are beyond the scope of this review.

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ANSWERS AND EXPLANATIONS TO PATIENT CASES

1. **Answer: B**

Given this patient's CAD, type 2 diabetes, dyslipidemia, gastroesophageal reflux disease, hypertension, obstructive sleep apnea, and ischemic cardiomyopathy, only Answer B consists of the three agents that reduce morbidity and mortality among these comorbidities: carvedilol (HF and potentially diabetes), spironolactone (HF progression post-MI), and lisinopril (HF and diabetes), making Answer B correct and Answers A (Amlodipine, clopidogrel, and sitagliptin.), C (Pravastatin, amlodipine, and aspirin 325 mg), and D (Prasugrel, sildenafil, and atenolol) incorrect.

2. **Answer: C**

Although many of these tests might be helpful in this patient's differential diagnosis, the most likely consideration would be respiratory decompensation or an HF decompensation, given the patient's presentation; thus, a chest radiograph, ECHO, BNP, and lactate would be most appropriate, making Answer C correct. Answer A is incorrect; although an infectious etiology cannot be ruled out, it is less likely than noninfectious cardiorespiratory differentials. Answer B is not the best answer; although serial troponins and an ABG could be argued for, liver function tests are unlikely to provide additional insight into this patient's differential. Answer D also is not the best answer; although it provides additional information for hemodynamic assessment, it is lacking in an assessment of cardiac function, which is a likely contributor to the patient's presentation.

3. **Answer: D**

Dobutamine would be the ideal choice, given the patient's acutely worsening cardiac index and malperfusion and its rapid onset; this would facilitate β_2 arterial vasodilation (both SVR and PVR reduction) as well as increase chronotropy and inotropy. Milrinone, although likely beneficial for both cardiac index and PVR reduction, would be less favorable, given that its time to peak effect will exceed 6 hours (the time required to achieve greater than 87.5% of steady state, even in normal renal function), and the dosing is rather high for safe initiation of therapy. Neither norepinephrine nor epinephrine would be favored (particularly at these doses) because the patient's MAP is currently elevated, and these agents would contribute to increased α_1-mediated afterload increases, causing increased myocardial workload in a patient who already has decompensated systolic HF. Answer D (dobutamine 5 mcg/kg/min) is correct, and Answers A (norepinephrine 0.08 mcg/kg/min), B (epinephrine 0.08 mcg/kg/min), and C (Milrinone 0.75 mcg/kg/min) are incorrect.

4. **Answer: D**

The patient's troponin elevation is most likely a result of the patient's decompensated HF (especially considering clinical evidence of acute HF, such as high BNP, CVP, and low cardiac output) superimposed on chronic CAD (Answer D is correct). Given the patient's chemistry and ABG results, chronic obstructive pulmonary disease (Answer B) is highly unlikely. Although this presentation could be attributed to NSTEMI (Answer A), it is less likely, given the lack of new ST-T changes on ECG and the accompanying evidence of an HF exacerbation. Sepsis (Answer C) is also a potential contributor to troponin elevation; by comparison, an infectious etiology is less likely in this patient's case.

5. **Answer: A**

Ticagrelor has been associated with adenosine-mediated dyspnea and bradycardia; therefore, this medication should be evaluated as a contributing cause. According to the current ACC/AHA guidelines, this patient should continue $P2Y_{12}$ inhibitor therapy because his stent was placed less than 12 months ago, and he should continue aspirin therapy indefinitely (Answer A is correct). Conversely, it would not be appropriate to stop ticagrelor without switching to an alternative $P2Y_{12}$ inhibitor (Answer B is incorrect). A likely cause has been identified (Answer C is incorrect), and the patient has no evidence of hyperkalemia (Answer D is incorrect).

6. **Answer: A**

Because of this patient's hypotension and ongoing cardiogenic shock, synchronized cardioversion would be preferred (Answer A is correct). Metoprolol (Answer B) would be unfavorable because of the patient's concurrent dobutamine use and hypotension; similarly, amiodarone 300 mg intravenous push (Answer D) could contribute to further hypotension. The primary of role of adenosine (Answer C) is to slow AV nodal conduction when patients are tachycardiac with a regular rhythm

(i.e., supraventricular tachycardia) to terminate the arrhythmia or to help differentiate atrial from ventricular arrhythmias.

7. **Answer: C**

The physician is considering heparin anticoagulation while the patient is in the ICU. Although valuable to consider for long-term anticoagulation, the HAS-BLED score is not validated for bleeding risk specific to heparin anticoagulation. Nonetheless, given the patient's CHA_2DS_2-VASc score (annual stroke risk of 6.7%), anticoagulation should be considered, making Answer C correct and Answer D ("anticoagulation is not warranted") incorrect. Answer A, "the patient's risk of bleeding is the same as his stroke risk," and Answer B, "the patient's risk of stroke exceeds his risk of bleeding," are incorrect statements because the foundation of these scores does not include risk of thromboembolism during ICU stay, nor do the scores assess risk of bleeding on heparin infusions.

8. **Answer: C**

Given his ongoing cardiogenic shock, this patient is unlikely to remain stable without intervention; thus, Answer D is incorrect. To help stabilize him, intra-aortic balloon counterpulsation (Answer C) would best facilitate selective afterload reduction during systole (increasing cardiac output in a patient with a low ejection fraction and severe mitral regurgitation) while providing augmented diastolic pressures. Venovenous ECMO (Answer B) is unlikely to help because this means of MCS depends on a functional LV and RV to provide forward blood flow. Venoarterial ECMO (Answer A) may stabilize the patient; however, it would increase afterload on the aortic valve, which could worsen his mitral regurgitation. Furthermore, this degree of MCS may be unwarranted at this time unless the patient develops progressive hypoxic cardiopulmonary failure.

ANSWERS AND EXPLANATIONS TO SELF-ASSESSMENT QUESTIONS

1. **Answer: C**
A diagnosis of cardiogenic shock is most likely, given the patient's history, presentation, ongoing vasopressor requirement, elevated BNP, and lack of infectious symptoms (Answer A is incorrect). Furthermore, this patient has positive troponins and ST-segment elevation in leads II, III, and aVF (inferior leads), making STEMI the primary diagnosis and reason for cardiogenic shock (Answer C is correct; Answer D is incorrect). The presence of hypotension and bradycardia are more likely consistent with complications of an inferior MI with possible RV involvement, as opposed to an acute exacerbation of HF. The patient also has a preserved LVEF and not systolic HF (Answer B is incorrect).

2. **Answer: D**
ST-segment elevation in leads II, III, and aVF (inferior) are most consistent with the right coronary artery (Answer D is correct). Answers A (left main coronary artery), B (left anterior descending artery), and C (left circumflex coronary artery) are incorrect.

3. **Answer: B**
According to the SOAP II investigation, dopamine is associated with increased rates of adverse events in patients with cardiogenic shock compared with norepinephrine (Answer B is correct; Answer A is incorrect). Milrinone (Answer C), although helpful in some patients with HF, would not be favored, given the patient's current hypotension and acute renal failure. Normal saline administration (Answer D) would likely be detrimental because of the patient's signs of fluid overload, including BNP elevation, hyponatremia, and rales on examination.

4. **Answer: A**
Lidocaine would be most favorable in patients with ischemia-mediated ventricular arrhythmias, and although the patient is not currently in VT, he is having persistent premature ventricular contractions (bigeminy), further increasing concern for ongoing ischemia and myocardial irritability (Answer A is correct). Metoprolol (Answer C) and diltiazem (Answer D) would be contraindicated because of ongoing cardiogenic shock and vasopressor requirements. Amiodarone 300 mg intravenous push (Answer B) is no longer appropriate because the patient has a pulse and blood pressure and is currently not in VT. Furthermore, rapid administration of amiodarone may lead to worsening hypotension.

5. **Answer: A**
The patient's heart catheterization was performed through the femoral artery, a vessel that is much more difficult to compress to facilitate hemostasis. Hematomas can occur at the access site; however, the most serious bleeding complication associated with this access site is a retroperitoneal bleed (Answer A is correct). Answer B (dissection/rupture), Answer C (stent thrombosis), and Answer D (papillary muscle rupture) are incorrect.

6. **Answer: C**
Vasopressin would be favored because, when administered at normal physiologic doses, it mediates predominant increases in SVR while minimally affecting the PVR (Answer C is correct). Phenylephrine (Answer B), however, will increase both PVR and SVR by the α_1-receptors. Given the patient's ongoing hypotension, low cardiac index, and rising CVP, inaction (Answer A) would be inappropriate, and additional volume administration (Answer D) would be detrimental in the setting of volume overload and RV failure.

7. **Answer: B**
Amiodarone boluses would be safest if administered slowly for 10 minutes to avoid additional hypotension, followed by a continuous infusion (Answer B is correct; Answer A is incorrect). Metoprolol (Answer C) and diltiazem (Answer D) would be contraindicated because of ongoing cardiogenic shock and vasopressor requirements.

8. **Answer: D**
Because of the diagnosis of acute MI, current quality measures would require initiation of or documentation to contraindications for each item except for ACE inhibitors/ARBs (Answer D) because the patient still has an LVEF greater than 40%. Answer A (aspirin contraindication), Answer B (statin contraindication), and C (β-blocker contraindication) are incorrect.

APPENDIX A – OVERVIEW OF ANTI-ARRHYTHMICS

Class	MOA	Drug (available dosage forms)	ECG Effects			Management Considerations and Pearls	Notable Drug Interactions	Defibrillation Threshold	TdP risk (%)
			PR	QRS	QT/QTc				
IA	Sodium channel blockade (intermediate potency)	Quinidine (PO and IV)	↑↓	↑	↑	Different formulations available (no exact conversions) α-Blockade may contribute to hypotension Prolonged half-life in CHF and hepatic dysfunction May require dose adjustments in renal and/or hepatic dysfunction Enhanced AV node conduction usually requires combination with AV node-blocking agent	CYP2D6 (inhibitor) CYP3A4 (substrate and inhibitor) QT-prolonging drugs	↑	2–8
		Procainamide (IV)	↑	↑	↑	May contribute to hypotension Metabolized by hepatic acetylation Active metabolite (NAPA) renally eliminated and has increased class III properties Dialyzable	QT-prolonging drugs	↑↓	1–10
		Disopyramide (PO)	↑↓	↑	↑	May require dose adjustments in renal and/or hepatic dysfunction Potent anticholinergic adverse effects May be used to treat hypertrophic cardiomyopathy	CYP3A4 (substrate) QT-prolonging drugs	↑	1–3
IB	Sodium channel blockade (low potency)	Lidocaine (IV)	N/A	N/A	N/A or ↓	Ventricular arrhythmias only Increased risk of toxicities in hepatic dysfunction, CHF, and elderly patients Enhanced efficacy in ischemic tissue Adverse effects include CNS depression, somnolence, tremors, seizures Lidocaine concentrations may be checked (total: 1.5-5 mcg/mL; free: 0.5-2 mcg/mL)	CYP3A4 (substrate)	↑	
		Mexiletine (PO)				Ventricular arrhythmias only Prolonged half-life in CHF and hepatic dysfunction Oral derivative of lidocaine	CYP2D6 (substrate) CYP1A2	↑	
IC	Sodium channel blockade (high potency)	Flecainide (PO)	↑	↑	N/A or ↑	Useful for AF/flutter in patients without structural heart disease Increased risk of mortality post-MI (Cardiac Arrhythmia Suppression Trial [CAST]) Avoid in CHF because of its potent negative inotropic effects Combine with AV node blocker to prevent rapid atrial flutter	CYP2D6	↑	
		Propafenone (PO)				Useful for AF/flutter in patients without structural heart disease Increased risk of mortality post-MI (CAST trial) Avoid in CHF because of its potent negative inotropic effects Combine with AV node blocker to prevent rapid atrial flutter SR and IR formulations not equivalent	First-pass metabolism CYP2D6, CYP3A4, CYP1A2 substrate	↑↓	

APPENDIX A – OVERVIEW OF ANTI-ARRHYTHMICS *(continued)*

II	β-Blockade	Ex. Carvedilol (PO) Labetalol (PO/IV) Metoprolol (PO/IV) Esmolol (IV)	N/A or ↑	N/A	N/A or ↓	Sinus bradycardia AV block Hypotension more likely with nonselective agents Continuous infusions of esmolol or labetalol may contribute to large amounts of fluid	Predominantly CYP2D6	N/A	
III	Potassium channel blockade	Dofetilide (PO)	N/A or ↑	N/A or ↑	↑	Requires ECG monitoring for initiation, dose titration, reinitiation, or introduction of new interacting agents Requires renal dosing adjustments Mg and K monitoring Circulation 2024;149:e1–156.	CYP3A4 Trimethoprim, verapamil, HCTZ, and many others contraindicated QT-prolonging drugs	↓	1–8
		Ibutilide (IV)				Only indicated for cardioversion of AF or enhancement of electrical cardioversion	QT-prolonging drugs	↓	1–8
		Sotalol (PO/IV)				Requires renal dosing adjustments Bradycardia Mg and K monitoring Circulation 2024;149:e1–156.	QT-prolonging drugs	↓	1–6
		Amiodarone (PO/IV)				Multi–channel-blocking properties Monitor liver, thyroid, and pulmonary function tests Average half-life 53 days; highly lipophilic; active metabolite Bradycardia and hypotension High protein binding	Inhibits CYP3A4, CYP2D6, CYP2C9 Adjust warfarin and digoxin by 50%	↑	< 1
		Dronedarone (PO)				Multi–channel-blocking properties Only approved for AF/flutter; less efficacious vs. amiodarone Less potential for organ-system toxicity vs. amiodarone Avoid in CHF, particularly NYHA class III/IV Bradycardia Liver function tests within first 6 mo of therapy	Inhibits CYP3A4, CYP2D6, CYP2C9		< 1
IV	Calcium channel blockade	Verapamil (PO/IV)	↑	N/A	N/A	Hepatic dosing Sinus bradycardia Negative inotrope – avoid in patients with systolic heart failure	CYP3A4 (substrate and inhibitor)	N/A	
		Diltiazem (PO/IV)				Decreased dosing in elderly patients and patients with hepatic dysfunction Bradycardia Negative inotrope - avoid in patients with systolic heart failure	CYP3A4 (substrate and inhibitor)	N/A	
Misc.	Others	Digoxin (PO/IV)	↑	N/A	N/A	Alternative for rate control of AF/flutter; also used for systolic heart failure Decreased dosing in elderly and patients with kidney dysfunction Loading dose may be given based on ideal body weight Not dialyzable Digoxin concentrations may be checked (AF/flutter: 0.8-2 ng/mL; heart failure: 0.5-0.9 ng/mL)	Many interactions		
		Adenosine (IV)	↑			Half-life < 10 s Used for acute treatment of AV node reentrant tachycardias Can help to distinguish atrial from ventricular arrhythmias			
		Magnesium (IV)	↑			Used to treat Torsades and arrhythmias associated with hypomagnesemia Adjunctive treatment for atrial fibrillation with rapid ventricular response to slow ventricular rate. Given with AV nodal blocking agent			

AF = atrial fibrillation; AV = atrioventricular; CHF = congestive heart failure; CNS = central nervous system; ECG = electrocardiogram; HCTZ = hydrochlorothiazide; IR = immediate release; IV = intravenous(ly); MI = myocardial infarction; MOA = mechanism of action; NAPA = N-acetylprocainamide; N/A = not applicable; NYHA = New York Heart Association; PO = oral(ly); SR = sustained release; TdP = Torsades de pointes